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Table of Contents

1.0 Introduction 4

1.1 Project Overview 4

1.2 Objectives of the Network Design 5

2.0 Entire Network Layout 6

2.1 Overall Network Topology (Packet Tracer Design) 6

2.2 Kuala Lumpur (HQ) Network Layout 8

2.3 Australia Branch Network Layout 9

2.4 Singapore Branch Network Layout 9

2.5 India Branch Network Layout 11

3.0 IPv4 Addressing Plan 12

3.1 Addressing Design Strategy (VLSM) 12

3.2 VLAN IP Address Allocation 12

3.3 WAN and Server Farm Public IP Addressing 14

4.0 LAN and WAN Verification 15

4.1 Basic Switch and Router Verification 15

4.1.1 Hostname and Device Identification 15

4.1.2 Console, VTY and Enable Password Verification 15

4.1.3 Password Encryption and Banner Verification 18

4.2 VLAN Verification 20

4.2.1 VLAN Creation and Naming 20

4.3 Trunk and Native VLAN Verification 26

4.3.1 Trunk Port Verification 26

4.3.2 Native VLAN Implementation 30

4.4 Inter-VLAN Routing Verification 31

4.5 DHCP Verification 33

4.5.1 DHCP Pool Verification per Site 33

4.5.2 DHCP Verification 40

4.6 WAN Routing Verification 41

4.6.1 WAN Topology Overview 41
4.6.2 OSPF Verification 41
4.6.3 Default Static Route Verification 47
4.6.4 Routing Verification 51
5.0 Proposed WLAN Architecture 56
5.1 WLAN Architecture 56
5.2 Wireless Security Protocol 57
5.3 Access Point and Client Summary 59
5.4 WLAN DHCP Verification 60
6.0 EtherChannel Implementation 61
6.1 EtherChannel Verification on Switches 61
7.0 HSRP Implementation 63
7.1 HSRP Design and Purpose 63
7.2 HSRP Failover Verification 65
8.0 Server Farm Verification 66
8.1 DNS Server 66
8.2 Web Server 68
8.3 FTP Server 69
10.0 Layer 2 Security Attacks and Verification 71
10.1 MAC Address Table Overflow Attack 71
10.2 DHCP Spoofing Attack 74
10.3 STP Attack 74
10.4 VLAN Hopping Attack 75
11.0 Security Recommendations 76
11.1 Honeypot Deployment 76
11.2 Bastion Host Implementation 77
11.3 Network Monitoring and Logging 78
12. Conclusion 79
13. References 79

1.0 Introduction

1.1 Project Overview

This project is to design and simulate a secure, scalable, and efficient enterprise network infrastructure for Alstom, supporting its operations across Kuala Lumpur (HQ), Australia, Singapore, and India. The proposed network modernization aims to enhance service availability, improve administrative control, and strengthen the overall security posture of the organization.

Using Cisco Packet Tracer as the simulation platform, a logical network topology is developed to evaluate the effectiveness of a structured VLAN-based design that enables efficient traffic segmentation across departments and geographical locations. The design incorporates both wired and wireless network architectures to support diverse operational requirements at each site while ensuring seamless inter-branch connectivity.

The network infrastructure further emphasizes resilience, scalability, and security through the adoption of industry-standard routing, redundancy, and Layer 2 protection mechanisms. Centralized server services are implemented to support essential enterprise operations, while secure routing and switching strategies ensure reliable communication between headquarters, branch offices, and the remote server farm. Overall, the proposed solution delivers a robust and future-ready network design that aligns with Alstom's operational and security requirements.

1.2 Objectives of the Network Design

The objective of this project is to design, implement, and evaluate a secure, scalable, and efficient enterprise network infrastructure for Alstom by applying Switching and Routing Essentials concepts within a real-world organisational scenario. The project aims to develop a comprehensive multi-site network topology that interconnects the Kuala Lumpur headquarters with branch offices in Australia, Singapore, and India, using Cisco Packet Tracer as the simulation platform to validate design feasibility and functionality.

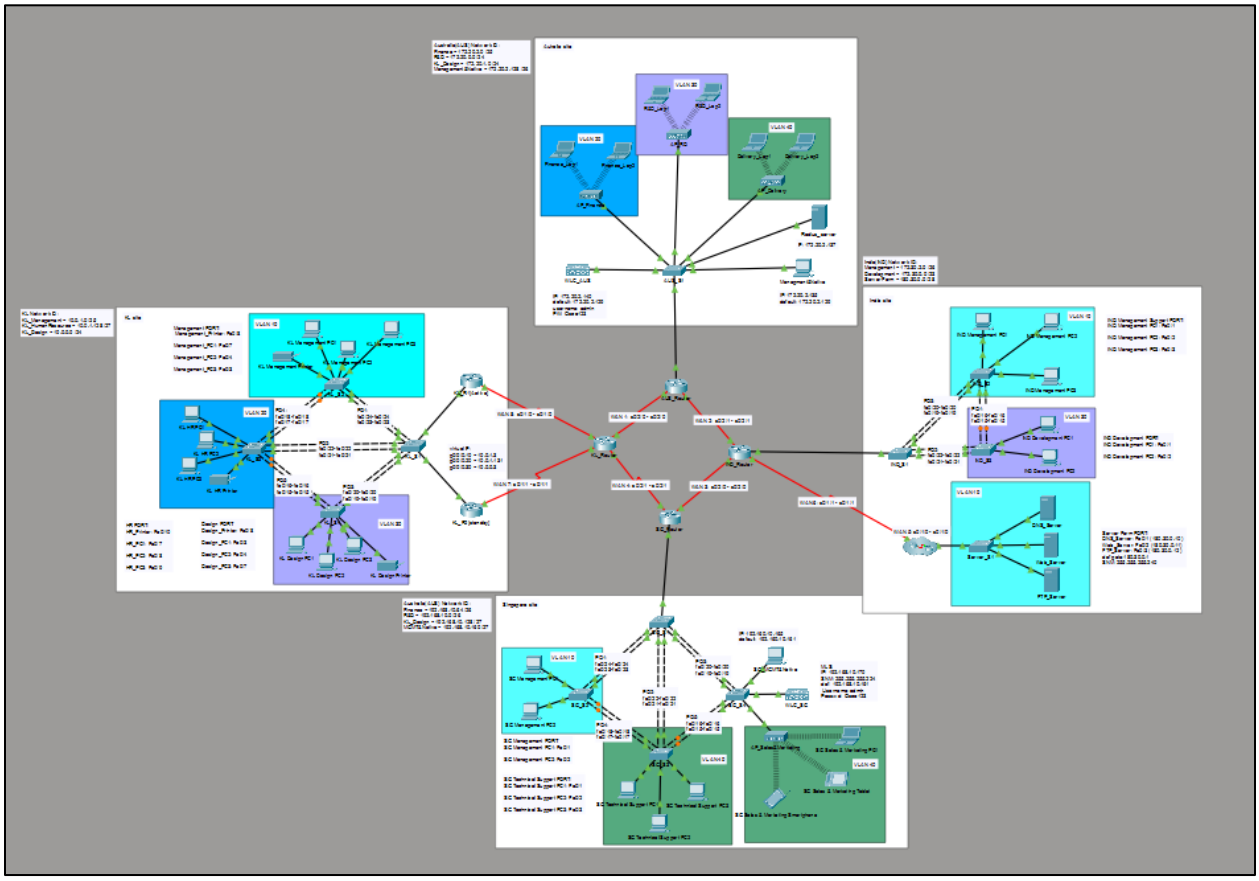
A key objective of this network design is to implement a structured Virtual Local Area Network (VLAN) architecture that enables effective traffic segmentation, improved performance, and enhanced administrative control across departments and geographical locations. In support of scalability and efficient resource utilization, the project also seeks to design and apply an optimized IPv4 addressing scheme using Variable Length Subnet Masking (VLSM), ensuring minimal address wastage while accommodating current and future expansion requirements.

Furthermore, the project aims to establish reliable inter-LAN and inter-WAN connectivity through the configuration of appropriate routing mechanisms, ensuring seamless communication between all sites. Network availability and resilience are emphasized through the deployment of redundancy and high-availability solutions to mitigate the impact of hardware or link failures. To support user mobility and operational flexibility, a centralized wireless network architecture is designed for selected branch offices using Wireless LAN Controllers (WLCs).

In addition, the project seeks to enhance network security by implementing Layer 2 security mechanisms to mitigate common LAN-based attacks and protect the integrity of the network infrastructure. Finally, essential server-based services are configured within a centralized server environment to support core enterprise operations, ensuring that the overall network design meets the operational, performance, and security requirements outlined in the given scenario.

2.0 Entire Network Layout

2.1 Overall Network Topology (Packet Tracer Design)

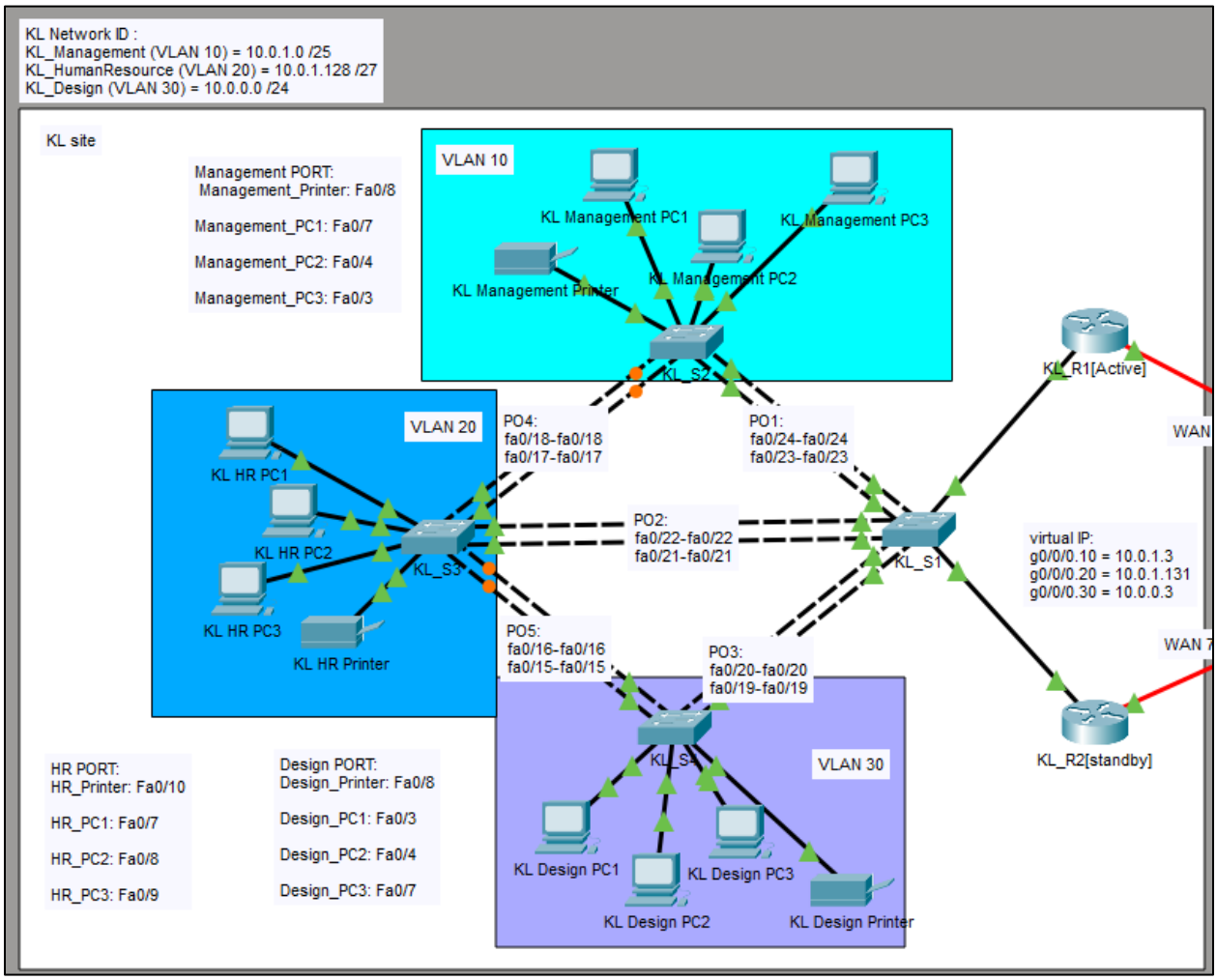


To enable smooth international operations, Alstom's architectural design centres on a highly resilient, hierarchical model spanning Kuala Lumpur (HQ), Australia, Singapore, and India. Multilayer switches serve as the main distribution point at the centre of each site, handling vital localized functions like DHCPv4 address assignment, inter-VLAN routing, and link aggregation via EtherChannel for higher performance. In order to improve internal privacy and optimize broadcast domains, the infrastructure uses a structured VLAN design to logically divide organizational traffic, particularly between departments like KL_Design, Finance, and the India Server Farm. Each branch has a range of end devices, such as printers and workstations, and the

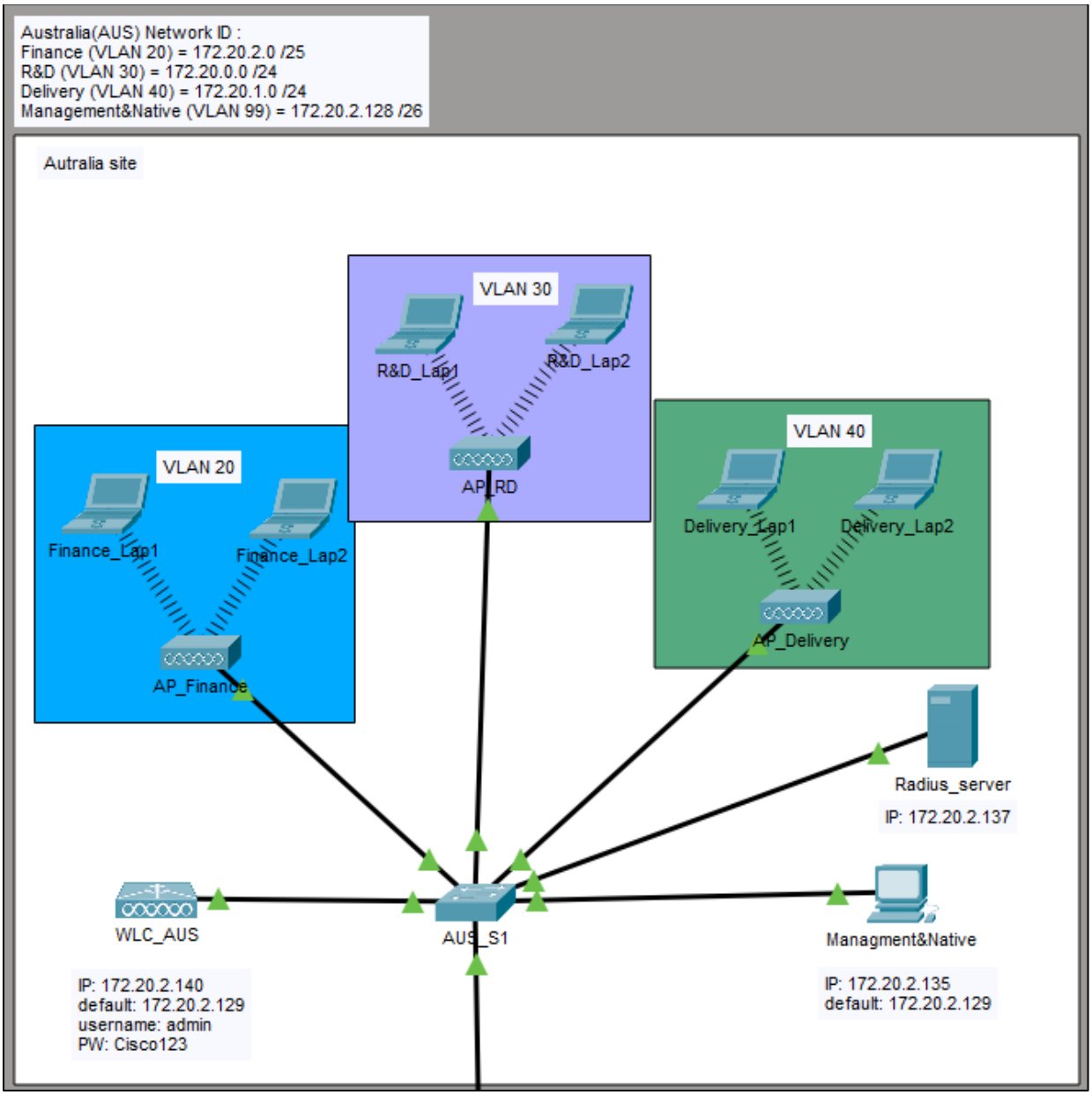
locations in Singapore and Australia incorporate a Wireless LAN Controller (WLC) to control laptop and smart device mobile connectivity.

This architecture is based on reliability and fault tolerance, especially in the Kuala Lumpur Headquarters and the link to distant subsidiaries. To guarantee that the network continues to function even in the case of a device or link failure, the architecture includes redundant physical links and gateway redundancy via the Hot Standby Router Protocol (HSRP). Using a combination of OSPF dynamic routing and default static routes, the wide area network (WAN) is organised around a central routing hub that links the KL site with the branches in Australia, Singapore, and India. This multi-path arrangement guarantees that traffic is automatically diverted to preserve ongoing communication in the event that a principal router or link goes unavailable. Additionally, all overseas departments have consistent access to DNS, FTP, and Web services thanks to the strategic integration of the centralised Server Farm in India into the global routing table.

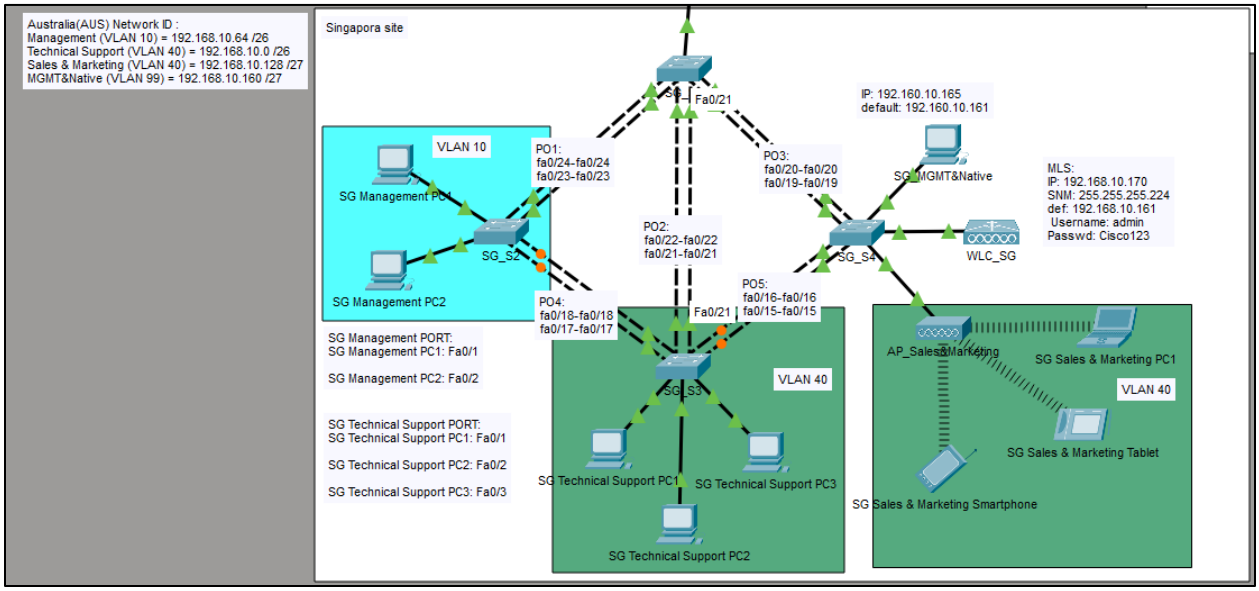
2.2 Kuala Lumpur (HQ) Network Layout



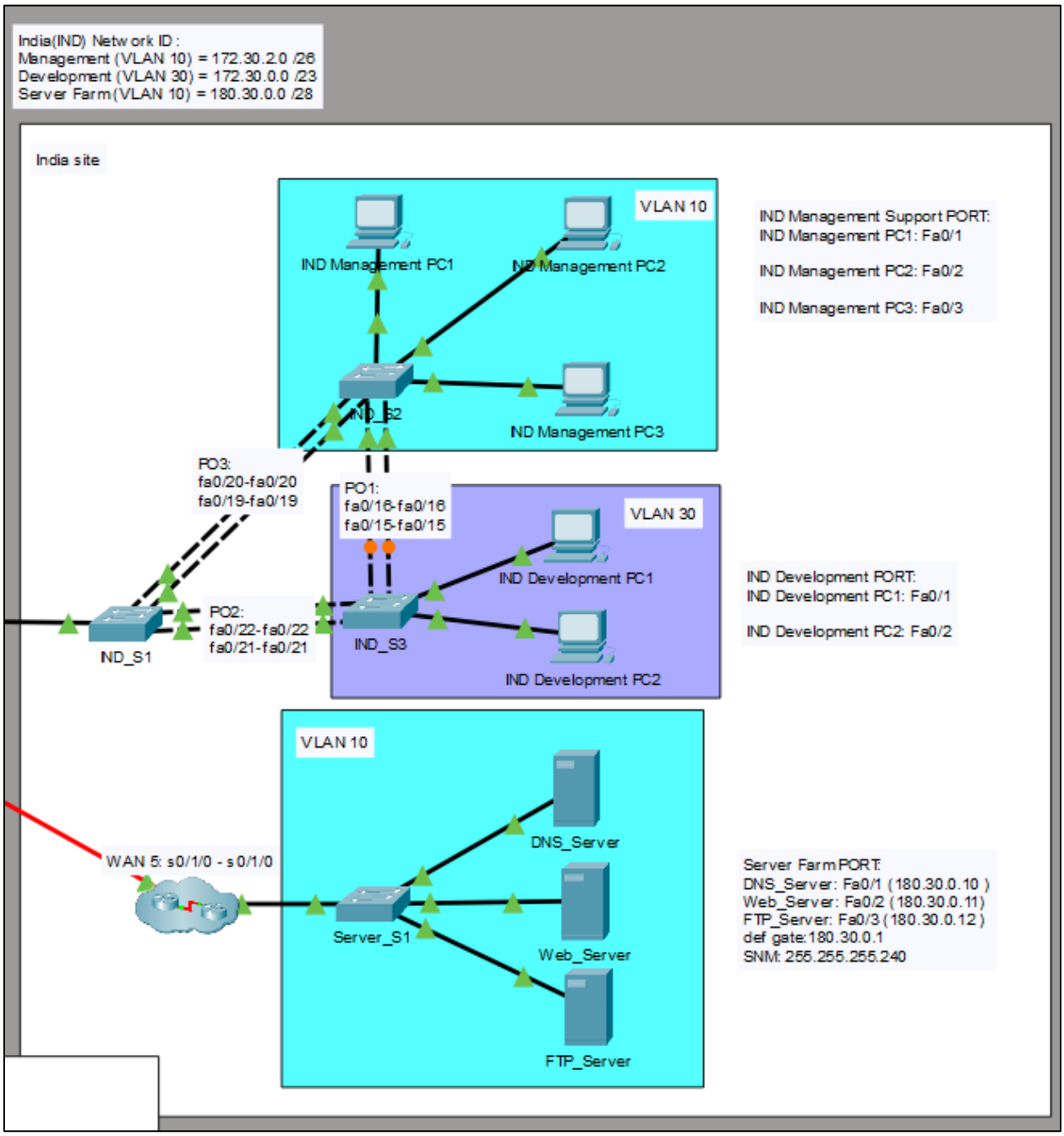
2.3 Australia Branch Network Layout



2.4 Singapore Branch Network Layout


Australia(AUS) Network ID :
Management (VLAN 10) = 192.168.10.64 /26
Technical Support (VLAN 40) = 192.168.10.0 /26
Sales & Marketing (VLAN 40) = 192.168.10.128 /27
MGMT&Native (VLAN 99) = 192.168.10.160 /27
Singapora site
Fa0/21
IP: 192.160.10.165
default: 192.160.10.161
VLAN 10
PO1:
fa0/24-fa0/24
fa0/23-fa0/23
PO3:
fa0/20-fa0/20
fa0/19-fa0/19
SG_MGMT&Native
MLS:
IP: 192.168.10.170
SNM: 255.255.255.224
def: 192.168.10.161
Username: admin
Passwd: Cisco123
SG Management PC1
PO2:
fa0/22-fa0/22
fa0/21-fa0/21
SG_S2
SG_S4
WLC_SG
SG Management PC2
PO4:
fa0/18-fa0/18
fa0/17-fa0/17
PO5:
fa0/16-fa0/16
fa0/15-fa0/15
Fa0/21
SG Management PORT:
SG Management PC1: Fa0/1
SG Management PC2: Fa0/2
AP_Sales&Marketing
SG Sales & Marketing PC1
VLAN 40
SG_S3
VLAN 40
SG Technical Support PORT:
SG Technical Support PC1: Fa0/1
SG Technical Support PC2: Fa0/2
SG Technical Support PC3: Fa0/3
SG Sales & Marketing Tablet
SG Technical Support PC1
SG Technical Support PC3
SG Sales & Marketing Smartphone
SG Technical Support PC2

2.5 India Branch Network Layout



3.0 IPv4 Addressing Plan

3.1 Addressing Design Strategy (VLSM)

The addressing design applies VLSM principles so subnet sizes can be aligned with the expected host counts per VLAN, improving utilisation and supporting scalable growth compared to fixed-length subnetting. Internal LAN VLANs are allocated from private address space, while public address space is reserved for WAN point-to-point links and the server farm network where required by the scenario constraints.

To maintain consistency, each VLAN's default gateway is implemented via an SVI (or an HSRP virtual IP where redundancy is used). Infrastructure addresses (e.g., gateways, switch management, WLC interfaces, and servers) are reserved and excluded from DHCP scopes to prevent accidental assignment to endpoints.

Area	Deparment	Network Information			Network ID	Subnet Mask	Default Gate Way	Range of Usable IP Address		Broadcast Addrerss
		Number of Hosts	Vlan ID	Vlan Name				First Usable IP Address	Last Usable IP Address	
KL Site	KL_Management	70	10	Central control	10.0.1.0 /25	255.255.255.128	10.0.1.1	10.0.1.1	10.0.1.126	10.0.1.127
	KL_HumanResource (HR)	30	20	Admin data	10.0.1.128 /27	255.255.255.224	10.0.1.129	10.0.1.129	10.0.1.158	10.0.1.159
	KL_Design	200	30	Innovation group	10.0.0.0 /24	255.255.255.0	10.0.0.1	10.0.0.1	10.0.0.254	10.0.0.255
	Native		99	Native	10.255.255.254					
	BlackHole	Unsed Port	999	BlackHole						
Australia	Finance	80	20	Admin data	172.20.2.0 /25	255.255.255.128	172.20.2.1	172.20.2.1	172.20.2.126	172.20.2.127
	R&D	150	30	Innovation group	172.20.0.0 /24	255.255.255.0	172.20.0.1	172.20.0.1	172.20.0.254	172.20.0.255
	Delivery	150	40	Customer-facing	172.20.1.0 /24	255.255.255.0	172.20.1.1	172.20.1.1	172.20.1.254	172.20.1.255
	Management&Native	50	99	Native	172.20.2.128 /26	255.255.255.192	172.20.2.129	172.20.2.129	172.20.2.190	172.20.2.191
	BlackHole	Unsed Port	999	BlackHole						
India	Management	50	10	Central control	172.30.2.0 /26	255.255.255.192	172.30.2.1	172.30.2.1	172.30.2.62	172.30.2.63
	Development	350	30	Innovation group	172.30.0.0 /23	255.255.254.0	172.30.0.1	172.30.0.1	172.30.1.254	172.30.1.255
	Server Farm	10	10	Central control	180.30.0.0 /28	255.255.255.240	180.30.0.1	180.30.0.1	180.30.0.14	180.30.0.15
	Native		99	Native	172.30.255.254					
	BlackHole	Unsed Port	999	BlackHole						
Singapora	Management	50	10	Central control	192.168.10.64 /26	255.255.255.192	192.168.10.65	192.168.10.65	192.168.10.126	192.168.10.127
	Technical Support	60	40	Customer-facing	192.168.10.0 /26	255.255.255.192	192.168.10.1	192.168.10.1	192.168.10.62	192.168.10.63
	Sales & Marketing	30	40	Customer-facing	192.168.10.128 /27	255.255.255.224	192.168.10.129	192.168.10.129	192.168.10.158	192.168.10.159
	MGMT&Native	25	99	Native	192.168.10.160 /27	255.255.255.224	192.168.10.161	192.168.10.161	192.168.10.190	192.168.10.191
	BlackHole	Unsed Port	999	BlackHole						

3.2 VLAN IP Address Allocation

<u>VLAN</u>

VLAN Table for SW_KL

Vlan ID	Deparment	Network ID	Subnet Mask
10	KL_Management	10.0.1.0 /25	255.255.255.128
20	KL_HumanResource (HR)	10.0.1.128 /27	255.255.255.224
30	KL_Design	10.0.0.0 /24	255.255.255.0
999	BlackHole		

VLAN Table for SW_AU

Vlan ID	Deparment	Network ID	Subnet Mask
20	Finance	172.20.2.0/25	255.255.255.128
30	R&D	172.20.0.0/24	255.255.255.0
40	Delivery	172.20.1.0/24	255.255.255.0
99	Management&Native	172.20.2.128/26	255.255.255.192
999	BlackHole		

VLAN Table for SW_India

Vlan ID	Deparment	Network ID	Subnet Mask
10	Management	172.30.2.0/26	255.255.255.192
30	Development	172.30.0.0/23	255.255.254.0
10	Server Farm	180.30.0.0/28	255.255.255.240
999	BlackHole		

VLAN Table for SW_SG

Vlan ID	Deparment	Network ID	Subnet Mask
10	Management	192.168.10.64/26	255.255.255.192
40	Technical Support	192.168.10.0/26	255.255.255.192
40	Sales&Marketing	192.168.10.128/27	255.255.255.224
99	MGMT&Native	192.168.10.160/27	255.255.255.224
999	BlackHole		

3.3 WAN and Server Farm Public IP Addressing

WAN IP Addressing

WAN	Network ID	Subnet Mask
WAN 1	200.100.100.0 /30	255.255.255.252
WAN 2	200.100.100.4 /30	255.255.255.252
WAN 3	200.100.100.8 /30	255.255.255.252
WAN 4	200.100.100.12 /30	255.255.255.252
WAN 5	200.100.100.16 /30	255.255.255.252
WAN 6	200.100.100.20 /30	255.255.255.252
WAN 7	200.100.100.24 /30	255.255.255.252
WAN 8	200.100.100.28 /30	255.255.255.252

Server Farm IP Addressing

Server Name	IP Address	Default Gateway

DNS_Server	180.30.0.10	180.30.0.1
Wed_Server	180.30.0.11	180.30.0.1
FTP_Server	180.30.0.12	180.30.0.1

4.0 LAN and WAN Verification

4.1 Basic Switch and Router Verification

4.1.1 Hostname and Device Identification

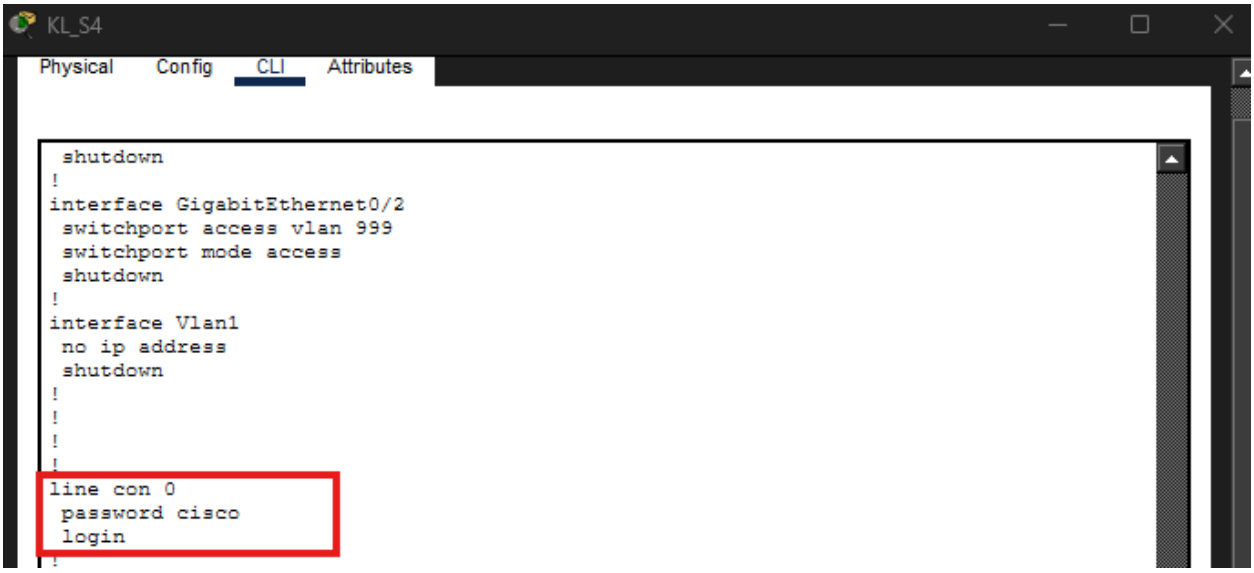
```
SG_S1
%LINK-5-CHANGED: Interface Port-channel2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to up

%LINK-5-CHANGED: Interface Port-channel3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel3, changed state to up

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up


SSG1>en
SSG1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SSG1(config)#hostname SSG1
```

This give the Router or switch a name as identification

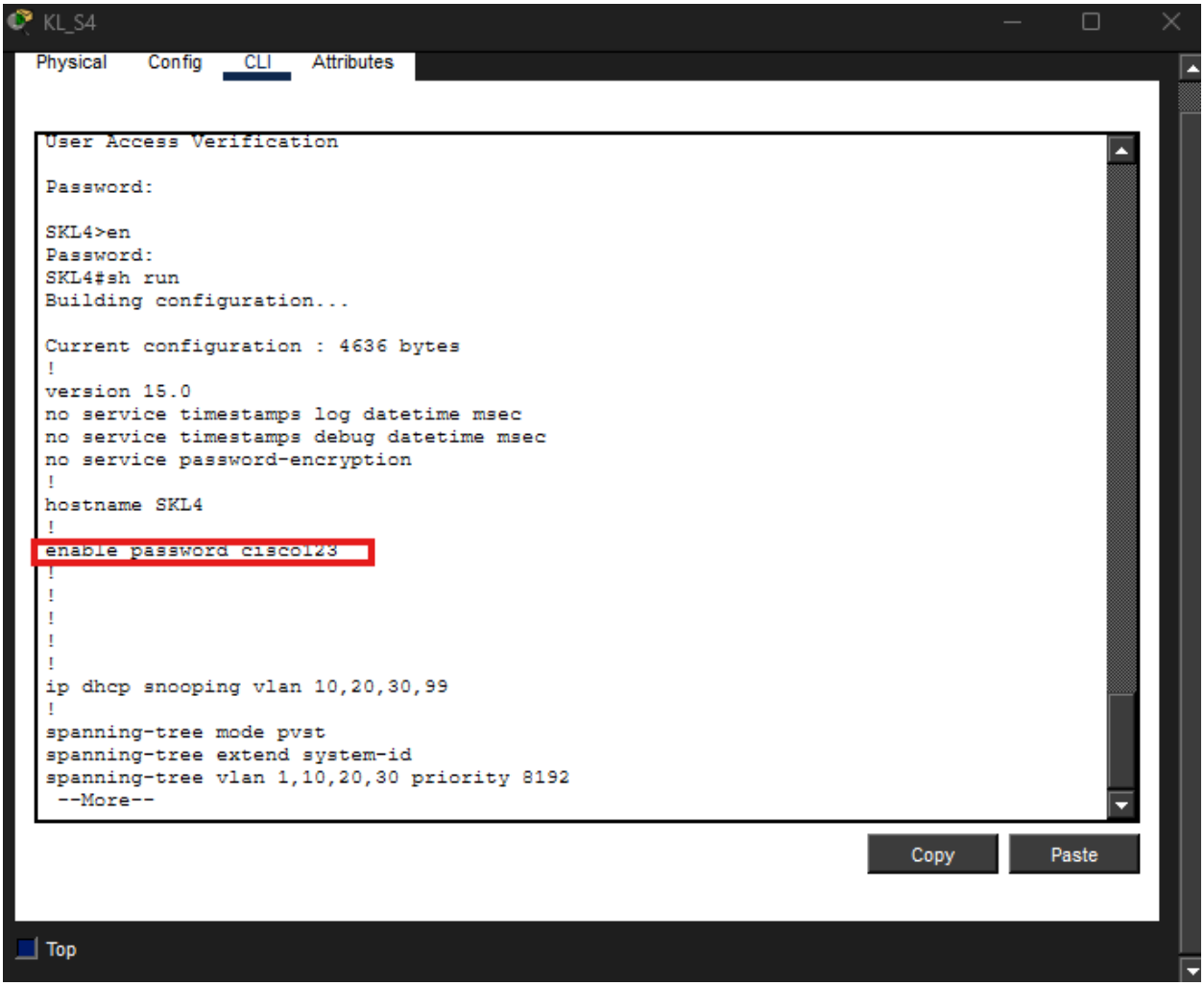
4.1.2 Console, VTY and Enable Password Verification

Administrative access is protected via console/VTY controls and privileged EXEC authentication to reduce unauthorised configuration changes.

Console line 0 password:



Enable Password:



Vty-password verification:

4.1.3 Password Encryption and Banner Verification

Password protection and warning banners are applied to reduce the risk of accidental disclosure and to provide an explicit notice of authorised-use requirements.

Banner-verification:

Password-Encryption-verification

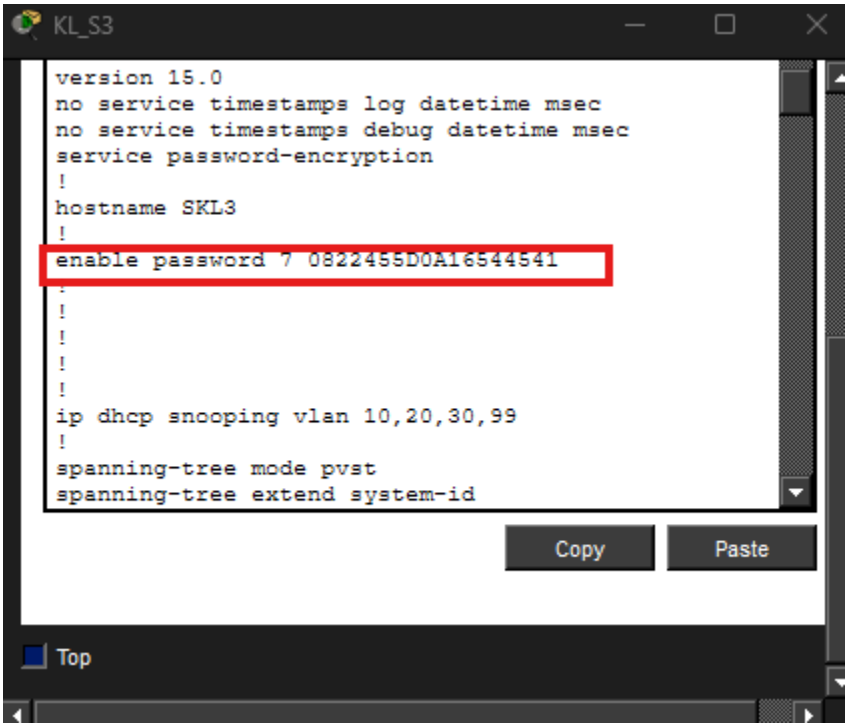
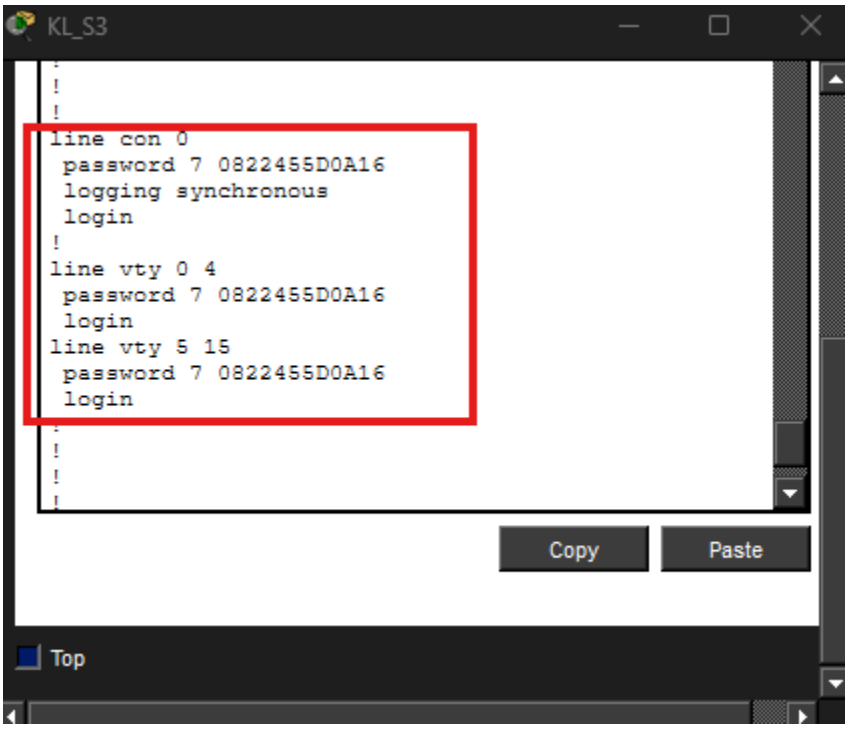


Figure 1 Password-Encryption



4.2 VLAN Verification

4.2.1 VLAN Creation and Naming

Vlan 10 is Central Control, the purpose of this vlan is to support core system management, monitoring, and control operations while ensuring high security and reliability.

Vlan 20 is Admin Data, the purpose of this vlan is to handle administrative and office-related data traffic, keeping internal records and staff communications isolated from other departments.

Vlan 30 is Innovation Group, the purpose of this vlan is to provide a flexible and isolated network environment for research, development, and testing activities without affecting production systems.

Vlan 40 is Customer-facing, the purpose of this vlan is to host services and applications that directly interact with customers, ensuring controlled access and protecting internal networks from external exposure.

Vlan 99 is Native, the purpose of this vlan is to carry untagged traffic across trunk links and to enhance network security by separating management and control traffic from user data VLANs.

Vlan 999 is blackhole, the purpose of this vlan is to store all the unused port, preventing unauthorized access.

KL Site

KL_S1:

```
SKL1#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   CentralControl                   active
20   AdminData                        active
30   InnovationGroup                  active
99   Native                           active
999  BlackHole                        active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SKL1#
```

KL_S2:

```
SKL2#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   CentralControl                   active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
20   AdminData                        active
30   InnovationGroup                  active
99   Native                           active
999  BlackHole                        active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SKL2#
```

KL_S3:

```
SKL3#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   CentralControl                   active
20   AdminData                        active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
30   InnovationGroup                  active
99   Native                           active
999  BlackHole                        active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/19, Fa0/20
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SKL3#
```

KL_S4:

```
SKL4#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   CentralControl                   active
20   AdminData                        active
30   InnovationGroup                  active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
99   Native                           active
999  BlackHole                        active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/17, Fa0/18
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SKL4#
```

<u>Australia Site</u>

AUS_S1:

```
SAU1#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
20   AdminData                        active
30   InnovationGroup                  active
40   CustomerFacing                   active
99   MGMT&Native                      active    Fa0/1, Fa0/2
999  BlackHole                        active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SAU1#
```

India Switch

IND_S1:

```
SIN1#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   CentralControl                   active
30   InnovationGroup                  active
99   Native                           active
999  BlackHole                        active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/23, Fa0/24
                                                Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SIN1#
```

IND_S2:

```
SIN2#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   CentralControl                   active    Fa0/1, Fa0/2, Fa0/3
30   InnovationGroup                  active
99   Native                           active
999  BlackHole                        active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/17
                                                Fa0/18, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SIN2#
```

IND_S3:

```
SIN3#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   CentralControl                   active
30   InnovationGroup                  active    Fa0/1, Fa0/2
99   Native                           active
999  Blackhole                        active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SIN3#
```

Server_S1:

```
Server_S1#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   CentralControl                   active    Fa0/1, Fa0/2, Fa0/3
30   InnovationGroup                  active
99   Native                           active
999  BlackHole                        active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Server_S1#
```

Singapore Switch

SG_S1:

```
SSG1#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   CentralControl                   active
40   Customer-facing                  active
99   native                           active
999  BlackHole                        active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SSG1#
```

SG_S2:

```
SSG2#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   CentralControl                   active    Fa0/1, Fa0/2
40   Customer-facing                  active
99   native                           active
999  BlackHole                        active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SSG2#
```

SG_S3:

```
SSG3#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   CentralControl                   active
40   Customer-facing                  active    Fa0/1, Fa0/2, Fa0/3
99   native                           active
999  BlackHole                        active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/19
                                                Fa0/20, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SSG3#
```

SG_S4:

```
SSG4#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   CentralControl                   active
40   Customer-facing                  active
99   native                           active    Fa0/1
999  BlackHole                        active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/17, Fa0/18, Fa0/21
                                                Fa0/22, Fa0/23, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SSG4#
```

4.3 Trunk and Native VLAN Verification

4.3.1 Trunk Port Verification

KL Switch

```
KL_S1
SKL1#
SKL1#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      99
Po2         on           802.1q         trunking      99
Po3         on           802.1q         trunking      99
Gig0/1      on           802.1q         trunking      99
Gig0/2      on           802.1q         trunking      99

Port        Vlans allowed on trunk
Po1         10,20,30,99
Po2         10,20,30,99
Po3         10,20,30,99
Gig0/1      10,20,30,99
Gig0/2      10,20,30,99

Port        Vlans allowed and active in management domain
Po1         10,20,30,99
Po2         10,20,30,99
Po3         10,20,30,99
Gig0/1      10,20,30,99
Gig0/2      10,20,30,99

Port        Vlans in spanning tree forwarding state and not pruned

SKL1#
Copy   Paste
```

Trunk Ports: Five interfaces are active trunks: Po1, Po2, Po3, Gig0/1, and Gig0/2.

Encapsulation: All trunks use 802.1q, which is the industry standard for tagging VLAN traffic.

Native VLAN: VLAN 99 is implemented as the native VLAN. This means any traffic on VLAN 99 is sent without a tag.

Allowed VLANs: Traffic is restricted to VLANs 10, 20, 30, and 99. All other VLANs are blocked from crossing these links.

AUS Switch

```
AUS_S1

SAU1>en
SAU1#sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/22      on           802.1q         trunking      99
Fa0/23      on           802.1q         trunking      99
Fa0/24      on           802.1q         trunking      99
Gig0/1      on           802.1q         trunking      99
Gig0/2      on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/22      20,30,40,99
Fa0/23      20,30,40,99
Fa0/24      20,30,40,99
Gig0/1      20,30,40,99
Gig0/2      20,30,40,99

Port        Vlans allowed and active in management domain
Fa0/22      20,30,40,99
Fa0/23      20,30,40,99
Fa0/24      20,30,40,99
Gig0/1      20,30,40,99
Gig0/2      20,30,40,99

Port        Vlans in spanning tree forwarding state and not pruned
 --More--
```

Copy Paste

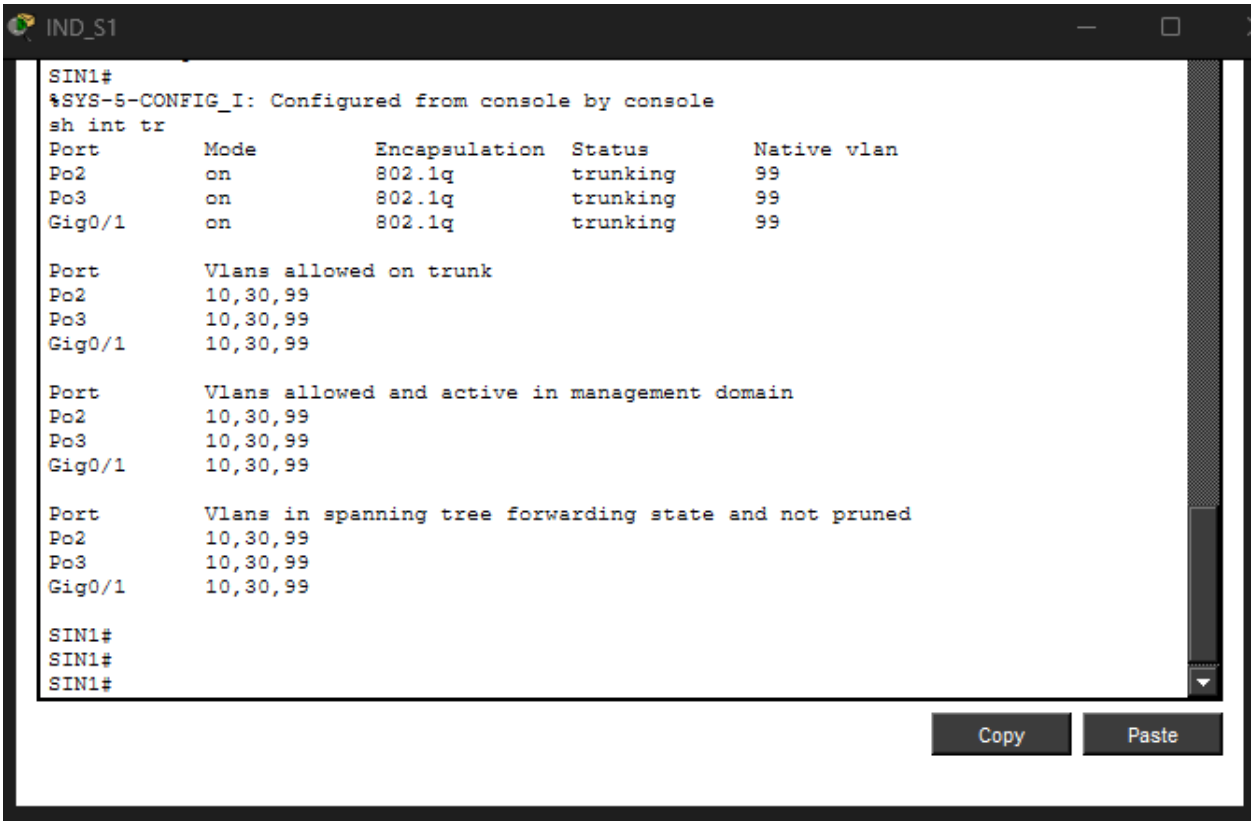
Trunk Ports: Five interfaces are active trunks: Fa0/22, Fa0/23, Fa0/24, Gig0/1, and Gig0/2.

Encapsulation: All trunks use 802.1q, which is the industry standard for tagging VLAN traffic.

Native VLAN: VLAN 99 is implemented as the native VLAN. This means any traffic on VLAN 99 is sent without a tag.

Allowed VLANs: Traffic is restricted to VLANs 20, 30, 40, and 99. All other VLANs are blocked from crossing these links.

IND_Switch



```
IND_S1
SIN1#
%SYS-5-CONFIG_I: Configured from console by console
sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Po2         on           802.1q         trunking      99
Po3         on           802.1q         trunking      99
Gig0/1      on           802.1q         trunking      99

Port        Vlans allowed on trunk
Po2         10,30,99
Po3         10,30,99
Gig0/1      10,30,99

Port        Vlans allowed and active in management domain
Po2         10,30,99
Po3         10,30,99
Gig0/1      10,30,99

Port        Vlans in spanning tree forwarding state and not pruned
Po2         10,30,99
Po3         10,30,99
Gig0/1      10,30,99

SIN1#
SIN1#
SIN1#
Copy
Paste
```

Trunk Ports: Three interfaces are active trunks: Po2, Po3 and Gig0/1

Encapsulation: All trunks use 802.1q, which is the industry standard for tagging VLAN traffic.

Native VLAN: VLAN 99 is implemented as the native VLAN. This means any traffic on VLAN 99 is sent without a tag.

Allowed VLANs: Traffic is restricted to VLANs 10, 30 and 99. All other VLANs are blocked from crossing these links.

SG Switch

```
SG_S1
SSG1#
SSG1#sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      99
Po2         on           802.1q         trunking      99
Po3         on           802.1q         trunking      99
Gig0/1      on           802.1q         trunking      99

Port        Vlans allowed on trunk
Po1         10,40,99
Po2         10,40,99
Po3         10,40,99
Gig0/1      10,40,99

Port        Vlans allowed and active in management domain
Po1         10,40,99
Po2         10,40,99
Po3         10,40,99
Gig0/1      10,40,99

Port        Vlans in spanning tree forwarding state and not pruned
Po1         10,40,99
Po2         10,40,99
Po3         10,40,99
Gig0/1      10,40,99
 --More--
Copy  Paste
```

Trunk Ports: Four interfaces are active trunks: Po1, Po2, Po3 and Gig0/1

Encapsulation: All trunks use 802.1q, which is the industry standard for tagging VLAN traffic.

Native VLAN: VLAN 99 is implemented as the native VLAN. This means any traffic on VLAN 99 is sent without a tag.

Allowed VLANs: Traffic is restricted to VLANs 10, 40 and 99. All other VLANs are blocked from crossing these links.

4.3.2 Native VLAN Implementation

KL Switch

```
SKL1#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      99
Po2         on           802.1q         trunking      99
Po3         on           802.1q         trunking      99
Gig0/1      on           802.1q         trunking      99
Gig0/2      on           802.1q         trunking      99
```

AUS Switch

```
SAU1#sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/22      on           802.1q         trunking      99
Fa0/23      on           802.1q         trunking      99
Fa0/24      on           802.1q         trunking      99
Gig0/1      on           802.1q         trunking      99
Gig0/2      on           802.1q         trunking      99
```

IND Switch

```
sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Po2         on           802.1q         trunking      99
Po3         on           802.1q         trunking      99
Gig0/1      on           802.1q         trunking      99
```

SG Switch

```
SSG1#sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      99
Po2         on           802.1q         trunking      99
Po3         on           802.1q         trunking      99
Gig0/1      on           802.1q         trunking      99
```

4.4 Inter-VLAN Routing Verification

KL router 1

```
RKL1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES unset  up                    up
GigabitEthernet0/0/0.1010.0.1.1        YES manual up                    up
GigabitEthernet0/0/0.2010.0.1.129      YES manual up                    up
GigabitEthernet0/0/0.3010.0.0.1        YES manual up                    up
GigabitEthernet0/0/1   unassigned      YES unset  administratively down down
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Serial0/1/0            200.100.100.29  YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
Serial0/2/0            unassigned      YES unset  administratively down down
Serial0/2/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
RKL1#
```

Kl router 2

KL_R2[standby]

```
RKL2#sh ip int bri
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES unset  up                    up
GigabitEthernet0/0/0.1010.0.1.2        YES manual up                    up
GigabitEthernet0/0/0.2010.0.1.130      YES manual up                    up
GigabitEthernet0/0/0.3010.0.0.2        YES manual up                    up
GigabitEthernet0/0/1   unassigned      YES unset  administratively down down
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Serial0/1/0            unassigned      YES unset  administratively down down
Serial0/1/1            200.100.100.25  YES manual up                    up
Serial0/2/0            unassigned      YES unset  administratively down down
Serial0/2/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

AUS router

AUS_Router

```
Main_RAUS>en
Main_RAUS#sh ip int bri
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES NVRAM  administratively down down
GigabitEthernet0/0/1   unassigned      YES NVRAM  up                    up
GigabitEthernet0/0/1.20172.20.2.1      YES manual up                    up
GigabitEthernet0/0/1.30172.20.0.1      YES manual up                    up
GigabitEthernet0/0/1.40172.20.1.1      YES manual up                    up
GigabitEthernet0/0/1.99172.20.2.129    YES manual up                    up
Serial0/1/0            unassigned      YES NVRAM  down                  down
Serial0/1/1            unassigned      YES NVRAM  down                  down
Serial0/2/0            200.100.100.2   YES manual up                    up
Serial0/2/1            200.100.100.5   YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

IND_router

IND_Router

Physical Config CLI Attributes

```
Main_RIND>en
Main_RIND#sh ip int bri
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES unset  administratively down down
GigabitEthernet0/0/1   unassigned      YES unset  up                    up
GigabitEthernet0/0/1.10172.30.2.1      YES manual up                    up
GigabitEthernet0/0/1.30172.30.0.1      YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Serial0/1/0            unassigned      YES unset  down                  down
Serial0/1/1            200.100.100.21  YES manual up                    up
Serial0/2/0            200.100.100.10  YES manual up                    up
Serial0/2/1            200.100.100.6   YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

SG router

SG_Router

Physical Config CLI Attributes

```
Main_RSG#sh ip int bri
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES unset  administratively down down
GigabitEthernet0/0/1   unassigned      YES unset  up                    up
GigabitEthernet0/0/1.10192.168.10.65   YES manual up                    up
GigabitEthernet0/0/1.40192.168.10.1    YES manual up                    up
GigabitEthernet0/0/1.41192.168.10.129  YES manual up                    up
GigabitEthernet0/0/1.99192.168.10.161  YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Serial0/1/0            unassigned      YES unset  down                  down
Serial0/1/1            unassigned      YES unset  down                  down
Serial0/2/0            200.100.100.9   YES manual up                    up
Serial0/2/1            200.100.100.14  YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

4.5 DHCP Verification

4.5.1 DHCP Pool Verification per Site

Separate DHCP pools are created per VLAN, with exclusions reserved for infrastructure devices (gateway SVIs/HSRP virtual IPs, network management, WLC interfaces, and servers). This reduces operational risk and maintains deterministic addressing for critical hosts. Where DHCP security mechanisms are implemented, the configuration aligns to Cisco's guidance for DHCP-related security controls and snooping behavior.

KL Router

```
Main_RKL#sh r | s dhcp
ip dhcp excluded-address 10.0.1.1 10.0.1.10
ip dhcp excluded-address 10.0.1.129 10.0.1.138
ip dhcp excluded-address 10.0.0.1 10.0.0.10
ip dhcp pool VLAN_10
 network 10.0.1.0 255.255.255.128
 default-router 10.0.1.3
 dns-server 180.30.0.10
ip dhcp pool VLAN_20
 network 10.0.1.128 255.255.255.224
 default-router 10.0.1.131
 dns-server 180.30.0.10
ip dhcp pool VLAN_30
 network 10.0.0.0 255.255.255.0
 default-router 10.0.0.3
 dns-server 180.30.0.10
Main_RKL#
```

KL_Router

Physical | Config | CLI | Attributes

```
Main_RKL>show ip dhcp pool

Pool VLAN_10 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 126
 Leased addresses               : 3
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.0.1.1             10.0.1.1         - 10.0.1.126        3    / 3     / 126

Pool VLAN_20 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 30
 Leased addresses               : 3
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.0.1.129           10.0.1.129       - 10.0.1.158        3    / 3     / 30

Pool VLAN_30 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 3
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.0.0.1             10.0.0.1         - 10.0.0.254        3    / 3     / 254
Main_RKL>
```

Copy | Paste

AUS Router

```
Main_RAUS#sh r | s dhcp
ip dhcp excluded-address 172.20.0.1 172.20.0.10
ip dhcp excluded-address 172.20.2.1 172.20.2.10
ip dhcp excluded-address 172.20.1.1 172.20.1.10
ip dhcp pool VLAN_30
 network 172.20.0.0 255.255.255.0
 default-router 172.20.0.1
 dns-server 180.30.0.10
ip dhcp pool VLAN_20
 network 172.20.2.0 255.255.255.128
 default-router 172.20.2.1
 dns-server 180.30.0.10
ip dhcp pool VLAN_40
 network 172.20.1.0 255.255.255.0
 default-router 172.20.1.1
 dns-server 180.30.0.10
Main_RAUS#
```

AUS_Router

Physical | Config | CLI | Attributes

```
Main_RAUS#sh ip dhcp pool

Pool VLAN_30 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 172.20.0.1           172.20.0.1       - 172.20.0.254      2    / 3     / 254

Pool VLAN_20 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 126
 Leased addresses               : 2
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 172.20.2.1           172.20.2.1       - 172.20.2.126      2    / 3     / 126

Pool VLAN_40 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 172.20.1.1           172.20.1.1       - 172.20.1.254      2    / 3     / 254
Main_RAUS#
```

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IND Switch

```
Main_RIND#sh r | s dhcp
ip dhcp excluded-address 172.30.2.1 172.20.2.10
ip dhcp excluded-address 172.30.0.1 172.30.0.10
ip dhcp excluded-address 172.30.2.1
ip dhcp pool VLAN_10
 network 172.30.2.0 255.255.255.192
 default-router 172.30.2.1
 dns-server 180.30.0.10
ip dhcp pool VLAN_30
 network 172.30.0.0 255.255.254.0
 default-router 172.30.0.1
 dns-server 180.30.0.10
Main_RIND#
```

IND_Router

Physical Config CLI Attributes

```
Main_RIND>en
Main_RIND#sh ip dhcp pool

Pool VLAN_10 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 62
 Leased addresses               : 3
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 172.30.2.1           172.30.2.1       - 172.30.2.62       3    / 3     / 62

Pool VLAN_30 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 510
 Leased addresses               : 2
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 172.30.0.1           172.30.0.1       - 172.30.1.254      2    / 3     / 510
Main_RIND#
```

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SG Router

```
Main_RSG#sh r | s dhcp
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.10.65 192.168.10.74
ip dhcp excluded-address 192.168.10.129 192.168.10.138
ip dhcp pool VLAN_10
 network 192.168.10.64 255.255.255.192
 default-router 192.168.10.65
 dns-server 180.30.0.10
ip dhcp pool VLAN_40
 network 192.168.10.0 255.255.255.192
 default-router 192.168.10.1
 dns-server 180.30.0.10
ip dhcp pool VLAN_41
 network 192.168.10.128 255.255.255.224
 default-router 192.168.10.129
 dns-server 180.30.0.10
Main_RSG#
```

SG_Router

Physical Config CLI Attributes

```
Main_RSG>sh ip dhcp pool

Pool VLAN_10 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 62
 Leased addresses               : 2
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.10.65        192.168.10.65    - 192.168.10.126    2    / 3     / 62

Pool VLAN_40 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 62
 Leased addresses               : 6
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.10.1         192.168.10.1     - 192.168.10.62     6    / 3     / 62

Pool VLAN_41 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 30
 Leased addresses               : 0
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.10.129       192.168.10.129   - 192.168.10.158    0    / 3     / 30
Main_RSG>
```

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Top

There is vlan 40 and vlan 41 which the same vlan 40 but in a different subnet

4.5.2 DHCP Verification

KL Router

```
Main_RKL>en
Main_RKL#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
10.0.1.12        00D0.D340.8092           --                     Automatic
10.0.1.13        0060.2F74.1185           --                     Automatic
10.0.1.14        0000.0C10.A111           --                     Automatic
10.0.1.140       000A.41B9.1E06           --                     Automatic
10.0.1.139       000B.BE85.5080           --                     Automatic
10.0.1.141       00D0.9775.7831           --                     Automatic
10.0.0.11        0001.968C.C6AB           --                     Automatic
10.0.0.15        0040.0B50.6D06           --                     Automatic
10.0.0.13        0002.1726.EDE8           --                     Automatic
Main_RKL#
```

AUS Router

```
Main_RAUS>en
Main_RAUS#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
172.20.0.13      0006.2A1B.0C79           --                     Automatic
172.20.0.12      00E0.F7BE.52C8           --                     Automatic
172.20.2.12      00E0.A31A.221A           --                     Automatic
172.20.2.13      0009.7C77.3D69           --                     Automatic
172.20.1.11      00E0.B006.1AEA           --                     Automatic
172.20.1.14      000A.F341.0D67           --                     Automatic
Main_RAUS#
```

IND Router

```
Main_RIND>en
Main_RIND#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
172.30.2.2       0001.438A.B3BC           --                     Automatic
172.30.2.3       0003.E4C8.975E           --                     Automatic
172.30.2.4       0060.47AE.12A7           --                     Automatic
172.30.0.11      0030.F274.5869           --                     Automatic
172.30.0.12      00D0.D395.AD7E           --                     Automatic
Main_RIND#
```

SG Router

```
Main_RSG>en
Main_RSG#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.76    0060.47C0.8DD7           --                     Automatic
192.168.10.75    000A.F3A3.D3CE           --                     Automatic
192.168.10.11    000A.4191.8496           --                     Automatic
192.168.10.12    0001.631C.2367           --                     Automatic
192.168.10.13    0001.9783.5B6C           --                     Automatic
192.168.10.14    00D0.D394.B232           --                     Automatic
192.168.10.17    0002.16E9.9D80           --                     Automatic
192.168.10.19    0000.0CA3.92C0           --                     Automatic
Main_RSG#
```

DHCP operation is validated using DHCP pool and binding outputs as well as client IP configuration results, demonstrating that clients are receiving correct subnet parameters and DNS/gateway values.

4.6 WAN Routing Verification

4.6.1 WAN Topology Overview

The WAN interconnects the four sites and enables access to the India server farm services. Dynamic routing is used to reduce manual administrative overhead and to improve scalability as routes change over time.

4.6.2 OSPF Verification

OSPF is configured to advertise relevant LAN and WAN networks within the routing domain, enabling automatic route propagation and consistent reachability between sites.

KL Main Router

```
Main_RKL#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    200.100.100.28 0.0.0.3 area 0
    200.100.100.24 0.0.0.3 area 0
    200.100.100.0 0.0.0.3 area 0
    200.100.100.12 0.0.0.3 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:07:23
    1.1.1.2              110      00:06:53
    1.1.1.3              110      00:06:53
    2.2.2.1              110      00:07:19
    3.3.3.1              110      00:07:24
    3.3.3.2              110      00:07:24
    3.3.3.3              110      00:07:24
    4.4.4.1              110      00:07:24
  Distance: (default is 110)

Main_RKL#sh ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.2           0   FULL/  -        00:00:36    200.100.100.29  Serial0/1/0
2.2.2.1           0   FULL/  -        00:00:36    200.100.100.14  Serial0/2/1
4.4.4.1           0   FULL/  -        00:00:36    200.100.100.2   Serial0/2/0
Main_RKL#
```

KL Router 1

```
RKL1#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.2
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.1.0 0.0.0.127 area 0
    10.0.1.128 0.0.0.31 area 0
    10.0.0.0 0.0.0.255 area 0
    200.100.100.28 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet0/0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:27:08
    1.1.1.2              110      00:27:32
    1.1.1.3              110      00:27:32
    2.2.2.1              110      00:27:27
    3.3.3.1              110      00:27:32
    3.3.3.2              110      00:27:32
    3.3.3.3              110      00:27:32
    4.4.4.1              110      00:29:19
  Distance: (default is 110)

RKL1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.3           1   FULL/DR         00:00:31    10.0.1.2        GigabitEthernet0/0/0.10
1.1.1.3           1   FULL/DR         00:00:31    10.0.0.2        GigabitEthernet0/0/0.30
1.1.1.1           0   FULL/  -        00:00:32    200.100.100.30  Serial0/1/0
1.1.1.3           1   FULL/DR         00:00:31    10.0.1.130      GigabitEthernet0/0/0.20
RKL1#show ip route ospf
     172.30.0.0/16 is variably subnetted, 2 subnets, 2 masks
O       172.30.0.0 [110/193] via 200.100.100.30, 02:42:01, Serial0/1/0
O       172.30.2.0 [110/193] via 200.100.100.30, 02:42:01, Serial0/1/0
     180.30.0.0/28 is subnetted, 1 subnets
O       180.30.0.0 [110/321] via 200.100.100.30, 02:42:01, Serial0/1/0
     192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
O       192.168.10.0 [110/129] via 200.100.100.30, 02:42:01, Serial0/1/0
O       192.168.10.64 [110/129] via 200.100.100.30, 02:42:01, Serial0/1/0
O       192.168.10.128 [110/129] via 200.100.100.30, 02:42:01, Serial0/1/0
O       192.168.10.160 [110/129] via 200.100.100.30, 02:42:01, Serial0/1/0
     200.100.100.0/24 is variably subnetted, 7 subnets, 2 masks
O       200.100.100.8 [110/192] via 200.100.100.30, 02:42:01, Serial0/1/0
O       200.100.100.12 [110/128] via 200.100.100.30, 02:42:01, Serial0/1/0
O       200.100.100.16 [110/320] via 200.100.100.30, 02:42:01, Serial0/1/0
O       200.100.100.20 [110/256] via 200.100.100.30, 02:42:01, Serial0/1/0
O       200.100.100.24 [110/65] via 10.0.1.2, 02:41:41, GigabitEthernet0/0/0.10
                       [110/65] via 10.0.1.130, 02:41:41, GigabitEthernet0/0/0.20
                       [110/65] via 10.0.0.2, 02:41:41, GigabitEthernet0/0/0.30
```

KL Router 2

```
RKL2#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.3
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.1.0 0.0.0.127 area 0
    10.0.1.128 0.0.0.31 area 0
    10.0.0.0 0.0.0.255 area 0
    200.100.100.24 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet0/0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:28:56
    1.1.1.2              110      00:29:20
    1.1.1.3              110      00:29:20
    2.2.2.1              110      00:29:15
    3.3.3.1              110      00:29:20
    3.3.3.2              110      00:29:20
    3.3.3.3              110      00:29:20
    4.4.4.1              110      00:31:07
  Distance: (default is 110)
```

```
RKL2#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:37    200.100.100.26  Serial0/1/1
1.1.1.2           1   FULL/BDR        00:00:36    10.0.1.129      GigabitEthernet0/0/0.20
1.1.1.2           1   FULL/BDR        00:00:36    10.0.0.1        GigabitEthernet0/0/0.30
1.1.1.2           1   FULL/BDR        00:00:36    10.0.1.1        GigabitEthernet0/0/0.10
RKL2#show ip route ospf
     172.30.0.0/16 is variably subnetted, 2 subnets, 2 masks
O       172.30.0.0 [110/193] via 200.100.100.26, 00:30:57, Serial0/1/1
O       172.30.2.0 [110/193] via 200.100.100.26, 00:30:57, Serial0/1/1
     180.30.0.0/28 is subnetted, 1 subnets
O       180.30.0.0 [110/321] via 200.100.100.26, 00:30:57, Serial0/1/1
     192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
O       192.168.10.0 [110/129] via 200.100.100.26, 00:30:57, Serial0/1/1
O       192.168.10.64 [110/129] via 200.100.100.26, 00:30:57, Serial0/1/1
O       192.168.10.128 [110/129] via 200.100.100.26, 00:30:57, Serial0/1/1
O       192.168.10.160 [110/129] via 200.100.100.26, 00:30:57, Serial0/1/1
     200.100.100.0/24 is variably subnetted, 7 subnets, 2 masks
O       200.100.100.8 [110/192] via 200.100.100.26, 00:30:57, Serial0/1/1
O       200.100.100.12 [110/128] via 200.100.100.26, 00:30:57, Serial0/1/1
O       200.100.100.16 [110/320] via 200.100.100.26, 00:30:57, Serial0/1/1
O       200.100.100.20 [110/256] via 200.100.100.26, 00:30:57, Serial0/1/1
O       200.100.100.28 [110/65] via 10.0.1.1, 00:30:57, GigabitEthernet0/0/0.10
                       [110/65] via 10.0.1.129, 00:30:57, GigabitEthernet0/0/0.20
                       [110/65] via 10.0.0.1, 00:30:57, GigabitEthernet0/0/0.30
```

AUS Router

```
Main_RAUS#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 4.4.4.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    172.20.2.0 0.0.0.127 area 0
    172.20.0.0 0.0.0.255 area 0
    172.20.1.0 0.0.0.255 area 0
    172.20.2.128 0.0.0.63 area 0
    200.100.100.0 0.0.0.3 area 0
    200.100.100.4 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet0/0/1
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:09:28
    1.1.1.2              110      00:09:08
    1.1.1.3              110      00:09:08
    2.2.2.1              110      00:09:33
    3.3.3.1              110      00:09:38
    3.3.3.2              110      00:09:38
    3.3.3.3              110      00:09:38
    4.4.4.1              110      00:09:38
  Distance: (default is 110)
```

```
Main_RAUS#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:37    200.100.100.1   Serial0/2/0
Main_RAUS#show ip route ospf
     200.100.100.0/24 is variably subnetted, 10 subnets, 2 masks
O       200.100.100.8 [110/192] via 200.100.100.1, 00:10:31, Serial0/2/0
O       200.100.100.12 [110/128] via 200.100.100.1, 00:10:31, Serial0/2/0
O       200.100.100.16 [110/320] via 200.100.100.1, 00:10:31, Serial0/2/0
O       200.100.100.20 [110/256] via 200.100.100.1, 00:10:31, Serial0/2/0
O       200.100.100.24 [110/128] via 200.100.100.1, 00:10:31, Serial0/2/0
O       200.100.100.28 [110/128] via 200.100.100.1, 00:10:31, Serial0/2/0
```

IND Router

```
Main_RIND#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 3.3.3.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    172.30.2.0 0.0.0.63 area 0
    172.30.0.0 0.0.1.255 area 0
    200.100.100.8 0.0.0.3 area 0
    200.100.100.12 0.0.0.3 area 0
    200.100.100.20 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet0/0/1
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:12:19
    1.1.1.2              110      00:11:50
    1.1.1.3              110      00:11:50
    2.2.2.1              110      00:12:19
    3.3.3.1              110      00:12:20
    3.3.3.2              110      00:12:20
    3.3.3.3              110      00:12:20
    4.4.4.1              110      00:12:19
  Distance: (default is 110)

Main_RIND#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.1           0   FULL/  -        00:00:30    200.100.100.9   Serial0/2/0
3.3.3.2           0   FULL/  -        00:00:37    200.100.100.22  Serial0/1/1
Main_RIND#show ip route ospf
     180.30.0.0/28 is subnetted, 1 subnets
O       180.30.0.0 [110/129] via 200.100.100.22, 00:13:08, Serial0/1/1
     200.100.100.0/24 is variably subnetted, 11 subnets, 2 masks
O       200.100.100.0 [110/192] via 200.100.100.9, 00:13:08, Serial0/2/0
O       200.100.100.12 [110/128] via 200.100.100.9, 00:13:08, Serial0/2/0
O       200.100.100.16 [110/128] via 200.100.100.22, 00:13:08, Serial0/1/1
O       200.100.100.24 [110/192] via 200.100.100.9, 00:13:08, Serial0/2/0
O       200.100.100.28 [110/192] via 200.100.100.9, 00:13:08, Serial0/2/0
```

SG Router

```
Main_RSG#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 2.2.2.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.10.64 0.0.0.63 area 0
    192.168.10.0 0.0.0.63 area 0
    192.168.10.128 0.0.0.31 area 0
    192.168.10.160 0.0.0.31 area 0
    200.100.100.8 0.0.0.3 area 0
    200.100.100.12 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet0/0/1
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:14:42
    1.1.1.2              110      00:14:13
    1.1.1.3              110      00:14:13
    2.2.2.1              110      00:14:43
    3.3.3.1              110      00:14:43
    3.3.3.2              110      00:14:43
    3.3.3.3              110      00:14:43
    4.4.4.1              110      00:14:42
  Distance: (default is 110)


Main_RSG#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:32    200.100.100.13  Serial0/2/1
3.3.3.1           0   FULL/  -        00:00:33    200.100.100.10  Serial0/2/0
Main_RSG#show ip route ospf
     200.100.100.0/24 is variably subnetted, 10 subnets, 2 masks
O       200.100.100.0 [110/128] via 200.100.100.13, 00:15:26, Serial0/2/1
O       200.100.100.4 [110/192] via 200.100.100.13, 00:15:26, Serial0/2/1
O       200.100.100.16 [110/192] via 200.100.100.10, 00:15:26, Serial0/2/0
O       200.100.100.20 [110/128] via 200.100.100.10, 00:15:26, Serial0/2/0
O       200.100.100.24 [110/128] via 200.100.100.13, 00:15:26, Serial0/2/1
O       200.100.100.28 [110/128] via 200.100.100.13, 00:15:26, Serial0/2/1
```

4.6.3 Default Static Route Verification

Where required, a default route is configured and (if necessary) originated into OSPF so that routers have a defined route of last resort. This approach aligns with Cisco guidance for default route origination within OSPF designs.

KL Main Router

```
Main_RKL#show ip route static
     172.20.0.0/16 is variably subnetted, 4 subnets, 3 masks
S       172.20.0.0/24 [1/0] via 200.100.100.2
S       172.20.1.0/24 [1/0] via 200.100.100.2
S       172.20.2.0/25 [1/0] via 200.100.100.2
S       172.20.2.128/26 [1/0] via 200.100.100.2
     172.30.0.0/16 is variably subnetted, 2 subnets, 2 masks
S       172.30.0.0/23 [1/0] via 200.100.100.14
S       172.30.2.0/26 [1/0] via 200.100.100.14
     180.30.0.0/28 is subnetted, 1 subnets
S       180.30.0.0 [1/0] via 200.100.100.14
     192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
S       192.168.10.0/26 [1/0] via 200.100.100.14
S       192.168.10.64/26 [1/0] via 200.100.100.14
S       192.168.10.128/27 [1/0] via 200.100.100.14
S       192.168.10.160/27 [1/0] via 200.100.100.14

Main_RKL#
```

Recursive static routes have been used in KL_Router

KL Router 1

```
RKL1#show ip route static
S*   0.0.0.0/0 [1/0] via 200.100.100.30
```

The default route has been used in KL_R1[Active]

KL Router 2

```
RKL2>en
RKL2#show ip route static
S*   0.0.0.0/0 [1/0] via 200.100.100.26
```

The default route has been used in KL_R2[Standby]

Australia Router

```
Main_RAUS#sh ip route static
     10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
S       10.0.0.0/24 [1/0] via 200.100.100.1
S       10.0.1.0/25 [1/0] via 200.100.100.1
S       10.0.1.128/27 [1/0] via 200.100.100.1
     172.30.0.0/16 is variably subnetted, 2 subnets, 2 masks
S       172.30.0.0/23 [1/0] via 200.100.100.6
S       172.30.2.0/26 [1/0] via 200.100.100.6
     180.30.0.0/28 is subnetted, 1 subnets
S       180.30.0.0 [1/0] via 200.100.100.6
     192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
S       192.168.10.0/26 [1/0] via 200.100.100.6
S       192.168.10.64/26 [1/0] via 200.100.100.6
S       192.168.10.128/27 [1/0] via 200.100.100.6
S       192.168.10.160/27 [1/0] via 200.100.100.6

Main_RAUS#
```

Recursive static routes have been used in AUS_Router

India Router

```
Main_RIND#sh ip ro static
     10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
S       10.0.0.0/24 [1/0] via 200.100.100.5
S       10.0.1.0/25 [1/0] via 200.100.100.5
S       10.0.1.128/27 [1/0] via 200.100.100.5
     172.20.0.0/16 is variably subnetted, 4 subnets, 3 masks
S       172.20.0.0/24 [1/0] via 200.100.100.5
S       172.20.1.0/24 [1/0] via 200.100.100.5
S       172.20.2.0/25 [1/0] via 200.100.100.5
S       172.20.2.128/26 [1/0] via 200.100.100.5
     192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
S       192.168.10.0/26 [1/0] via 200.100.100.9
S       192.168.10.64/26 [1/0] via 200.100.100.9
S       192.168.10.128/27 [1/0] via 200.100.100.9
S       192.168.10.160/27 [1/0] via 200.100.100.9
S*   0.0.0.0/0 [1/0] via 200.100.100.22

Main_RIND#
```

Recursive static routes have been used in IND_Router

Singapore Router

```
Main_RSG#sh ip ro static
     10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
S       10.0.0.0/24 [1/0] via 200.100.100.13
S       10.0.1.0/25 [1/0] via 200.100.100.13
S       10.0.1.128/27 [1/0] via 200.100.100.13
     172.20.0.0/16 is variably subnetted, 4 subnets, 3 masks
S       172.20.0.0/24 [1/0] via 200.100.100.10
S       172.20.1.0/24 [1/0] via 200.100.100.10
S       172.20.2.0/25 [1/0] via 200.100.100.10
S       172.20.2.128/26 [1/0] via 200.100.100.10
     172.30.0.0/16 is variably subnetted, 2 subnets, 2 masks
S       172.30.0.0/23 [1/0] via 200.100.100.10
S       172.30.2.0/26 [1/0] via 200.100.100.10
     180.30.0.0/28 is subnetted, 1 subnets
S       180.30.0.0 [1/0] via 200.100.100.10

Main_RSG#
```

Recursive static routes have been used in SG_Router

4.6.4 Routing Verification

KL Site

KL to Australia:

```
C:\>tracert 172.20.0.12

Tracing route to 172.20.0.12 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      10.0.0.1
  2   0 ms      1 ms      1 ms      200.100.100.30
  3   2 ms      2 ms      3 ms      200.100.100.2
  4   11 ms     4 ms      13 ms     172.20.0.12

Trace complete.
```

The next hop from KL to Australia was 200.100.100.30

KL to India:

```
C:\>tracert 172.30.2.3

Tracing route to 172.30.2.3 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      10.0.0.1
  2   3 ms      1 ms      1 ms      200.100.100.30
  3   0 ms      0 ms      1 ms      200.100.100.14
  4   10 ms     10 ms     39 ms     200.100.100.6
  5   11 ms     11 ms     10 ms     172.30.2.3

Trace complete.
```

The next hop from KL to India was 200.100.100.30

KL to Singapore:

```
C:\>tracert 192.168.10.11

Tracing route to 192.168.10.11 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      10.0.0.1
  2   0 ms      1 ms      1 ms      200.100.100.30
  3   2 ms      1 ms      2 ms      200.100.100.14
  4   1 ms      0 ms      10 ms     192.168.10.11

Trace complete.
```

The next hop from KL to Singapore was 200.100.100.30

<u>Australia Site</u>

Australia to KL:

```
C:\>tracert 10.0.1.144

Tracing route to 10.0.1.144 over a maximum of 30 hops:

  1   10 ms     16 ms     8 ms      172.20.0.1
  2   5 ms      6 ms      3 ms      200.100.100.1
  3   8 ms      4 ms      11 ms     200.100.100.29
  4   6 ms      23 ms     7 ms      10.0.1.144

Trace complete.
```

The next hop from Australia to KL was 200.100.100.1

Australia to India:

```
C:\>tracert 172.30.2.2

Tracing route to 172.30.2.2 over a maximum of 30 hops:

  1   25 ms     5 ms      10 ms     172.20.0.1
  2   8 ms      8 ms      8 ms      200.100.100.6
  3   4 ms      8 ms      8 ms      172.30.2.2

Trace complete.
```

The next hop from Australia to India was 200.100.100.6

Australia to Singapore:

```
C:\>tracert 192.168.10.13

Tracing route to 192.168.10.13 over a maximum of 30 hops:

  1   16 ms     4 ms      21 ms     172.20.0.1
  2   11 ms     4 ms      10 ms     200.100.100.6
  3   8 ms      4 ms      10 ms     200.100.100.9
  4   19 ms     8 ms      14 ms     192.168.10.13

Trace complete.
```

The next hop from Australia to Singapore was 200.100.100.6

<u>India Site</u>

India to KL:

```
C:\>tracert 10.0.1.13

Tracing route to 10.0.1.13 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      172.30.2.1
  2   1 ms      1 ms      1 ms      200.100.100.5
  3   1 ms      2 ms      2 ms      200.100.100.13
  4   3 ms      2 ms      3 ms      200.100.100.25
  5   12 ms     1 ms      2 ms      10.0.1.13

Trace complete.
```

The next hop from India to KL was 200.100.100.5

India to Australia:

```
C:\>tracert 172.20.2.12

Tracing route to 172.20.2.12 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      172.30.2.1
  2   1 ms      1 ms      1 ms      200.100.100.5
  3   7 ms      14 ms     3 ms      172.20.2.12

Trace complete.
```

The next hop from India to Australia was 200.100.100.5

India to Singapore:

```
C:\>tracert 192.168.10.75

Tracing route to 192.168.10.75 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      172.30.2.1
  2    19 ms     0 ms      1 ms      200.100.100.9
  3    1 ms      1 ms      1 ms      192.168.10.75

Trace complete.
```

The next hop from India to Singapore was 200.100.100.9

<u>Singapore Site</u>

Singapore to KL:

```
C:\>tracert 10.0.1.13

Tracing route to 10.0.1.13 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.10.65
  2    1 ms      1 ms      1 ms      200.100.100.13
  3    2 ms      1 ms      1 ms      200.100.100.25
  4    1 ms      0 ms      10 ms     10.0.1.13

Trace complete.
```

The next hop from Singapore to KL was 200.100.100.13

Singapore to Australia:

```
C:\>tracert 172.20.2.12

Tracing route to 172.20.2.12 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.10.65
  2    20 ms     1 ms      1 ms      200.100.100.10
  3    2 ms      1 ms      2 ms      200.100.100.5
  4    8 ms      0 ms      2 ms      172.20.2.12

Trace complete.
```

The next hop from Singapore to Australia was 200.100.100.10

Singapore to India:

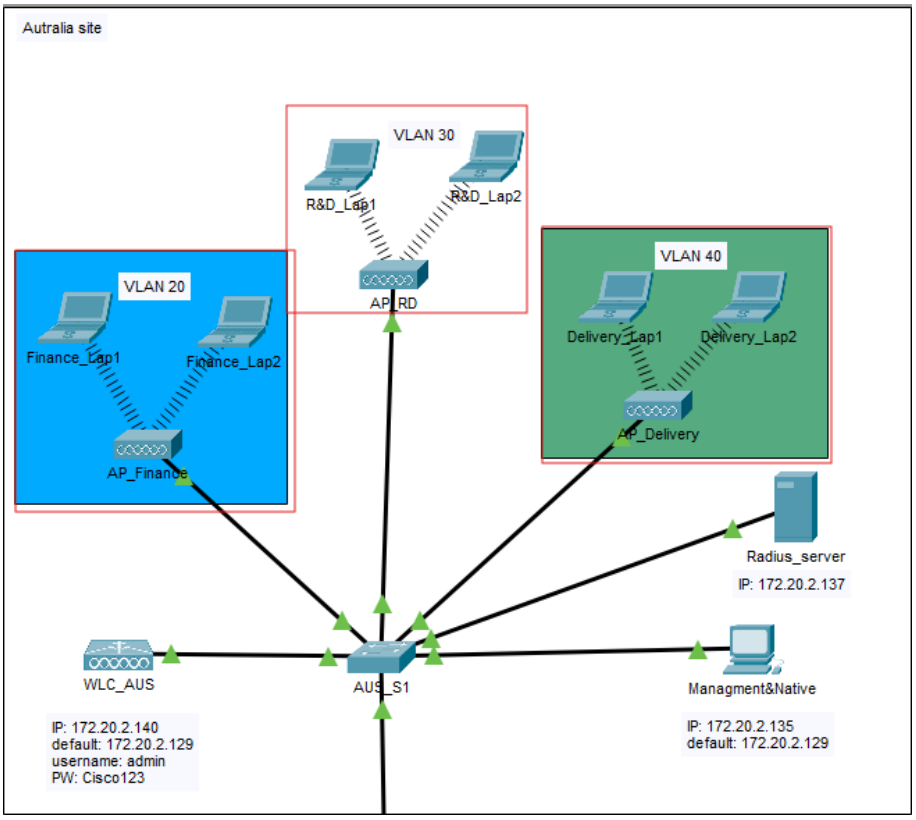
```
C:\>tracert 172.30.2.3

Tracing route to 172.30.2.3 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.10.65
  2    1 ms      1 ms      1 ms      200.100.100.10
  3    1 ms      11 ms     0 ms      172.30.2.3

Trace complete.
```

The next hop from Singapore to India was 200.100.100.10

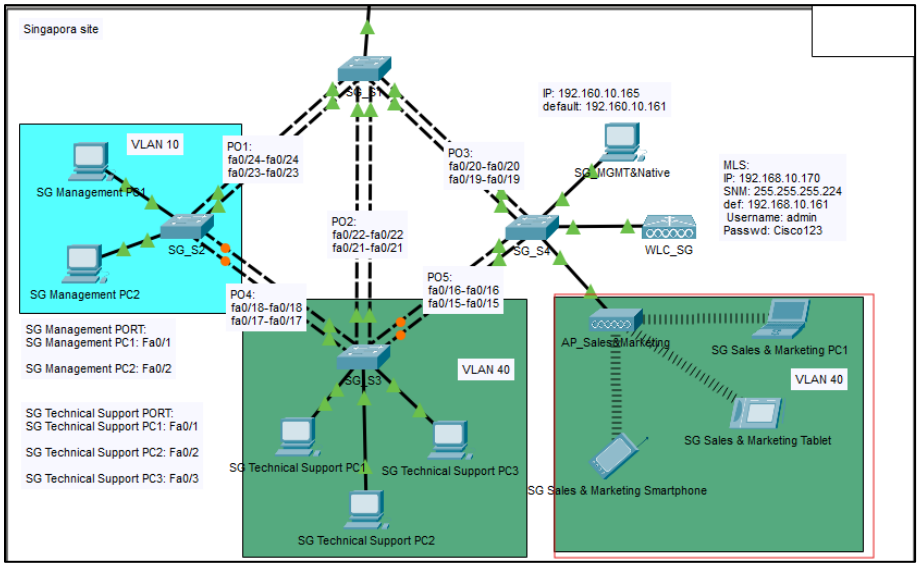
5.0 Proposed WLAN Architecture

5.1 WLAN Architecture

WLAN Architecture for Australia Site:

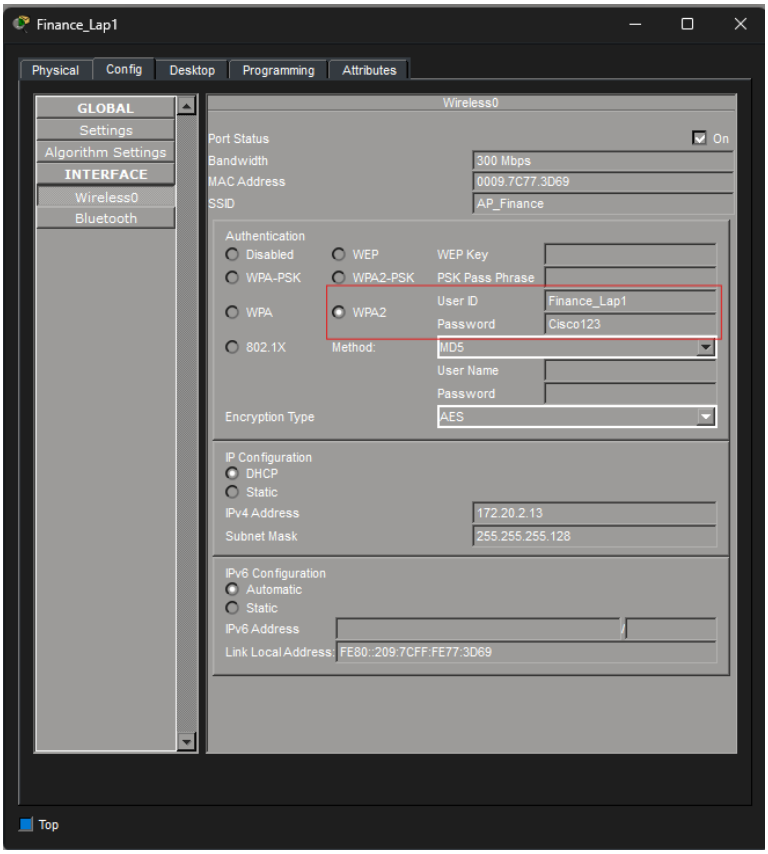


WLAN Architecture for Singapore Site:



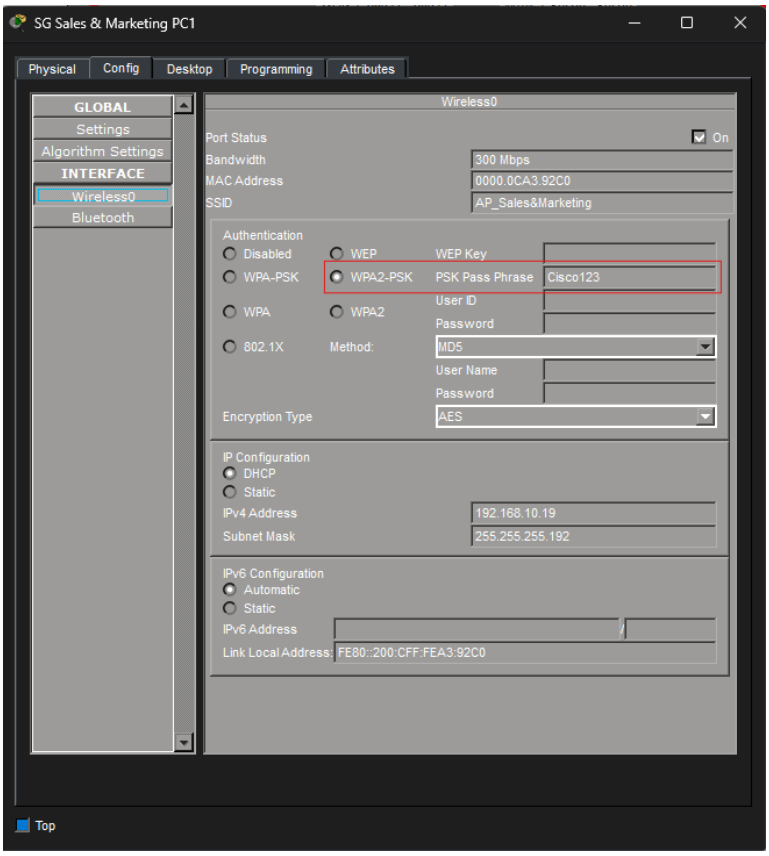
5.2 Wireless Security Protocol

Australia Site



Alstom plans to use WPA2 Enterprise in the Australian site to enable secure, centralized user authentication via a RADIUS server, ensuring strong encryption, per-user access control, and improved security for enterprise wireless networks.

Singapore Site



Alstom plans to utilize WPA2 Enterprise at the Singapore site to provide secure wireless access with AES encryption, while maintaining a simple and easy-to-manage WLAN design for a smaller branch environment.

5.3 Access Point and Client Summary

Australia Site

Access Point Summary

	Total	Up	Down	
802.11a/n/ac Radios	3	3	0	Detail
802.11b/g/n Radios	3	3	0	Detail
Dual-Band Radios	0	0	0	Detail
All APs	3	3	0	Detail

Client Summary

Current Clients	6	Detail
Excluded Clients	0	Detail
Disabled Clients	0	Detail

In this Site only 3 access points and 6 clients.

Singapore Site

Access Point Summary

	Total	Up	Down	
802.11a/n/ac Radios	1	1	0	Detail
802.11b/g/n Radios	1	1	0	Detail
Dual-Band Radios	0	0	0	Detail
All APs	1	1	0	Detail

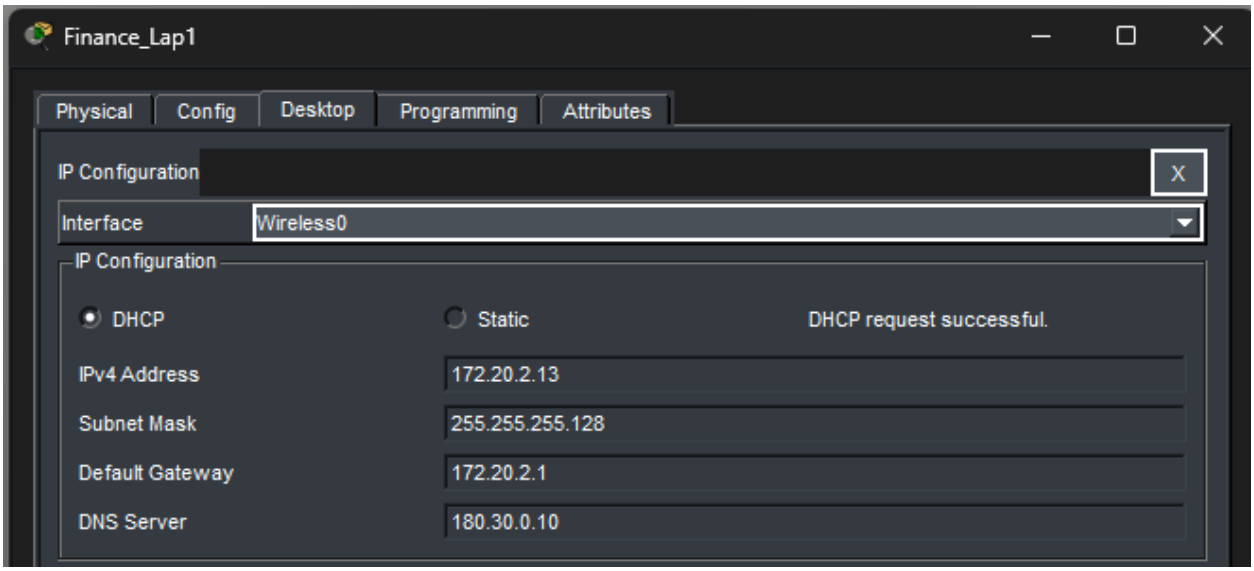
Client Summary

Current Clients	3	Detail
Excluded Clients	0	Detail
Disabled Clients	0	Detail

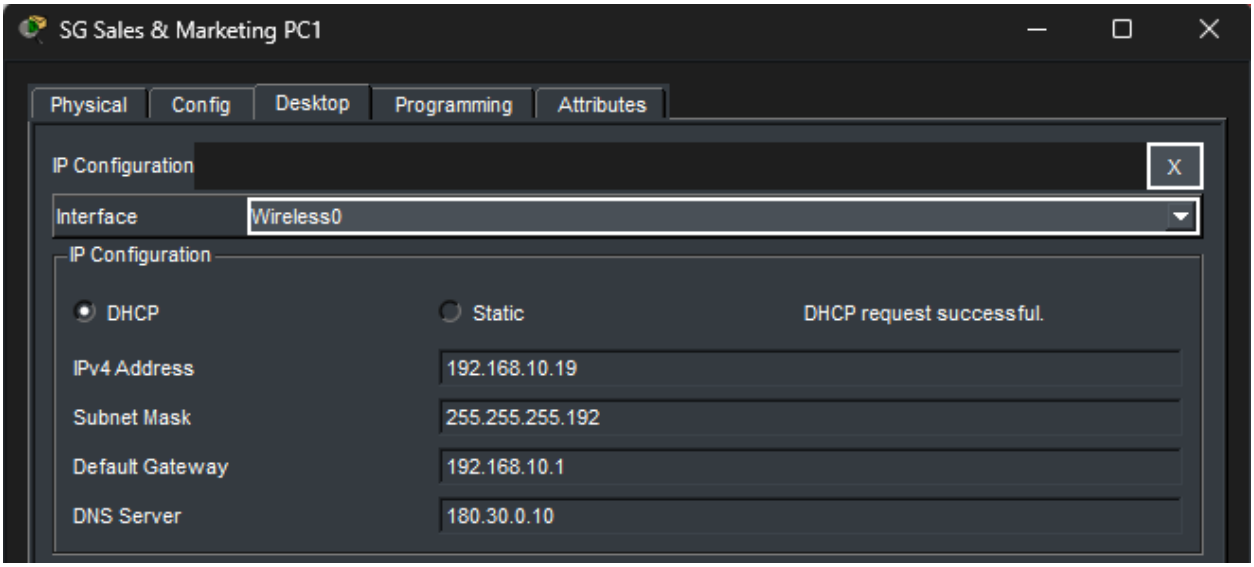
In this Site only 1 access points and 3 clients.

5.4 WLAN DHCP Verification

Australia Site



Singapore Site



6.0 EtherChannel Implementation

6.1 EtherChannel Verification on Switches

EtherChannel is configured between the required switching layers to support aggregated bandwidth and resilient uplink operation. Configuration consistency (speed/duplex and trunking parameters) is maintained across member interfaces to ensure stable channel operation.

KL Switch

KL_S1

Physical Config CLI Attributes

```
SKL1>en
SKL1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 3
Number of aggregators:           3

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           PAgP   Fa0/23(P) Fa0/24(P)
2      Po2(SU)           PAgP   Fa0/21(P) Fa0/22(P)
3      Po3(SU)           PAgP   Fa0/19(P) Fa0/20(P)
SKL1#
SKL1#
SKL1#
SKL1#
SKL1#
SKL1#
SKL1#
SKL1#
```

Copy Paste

IND_Switch

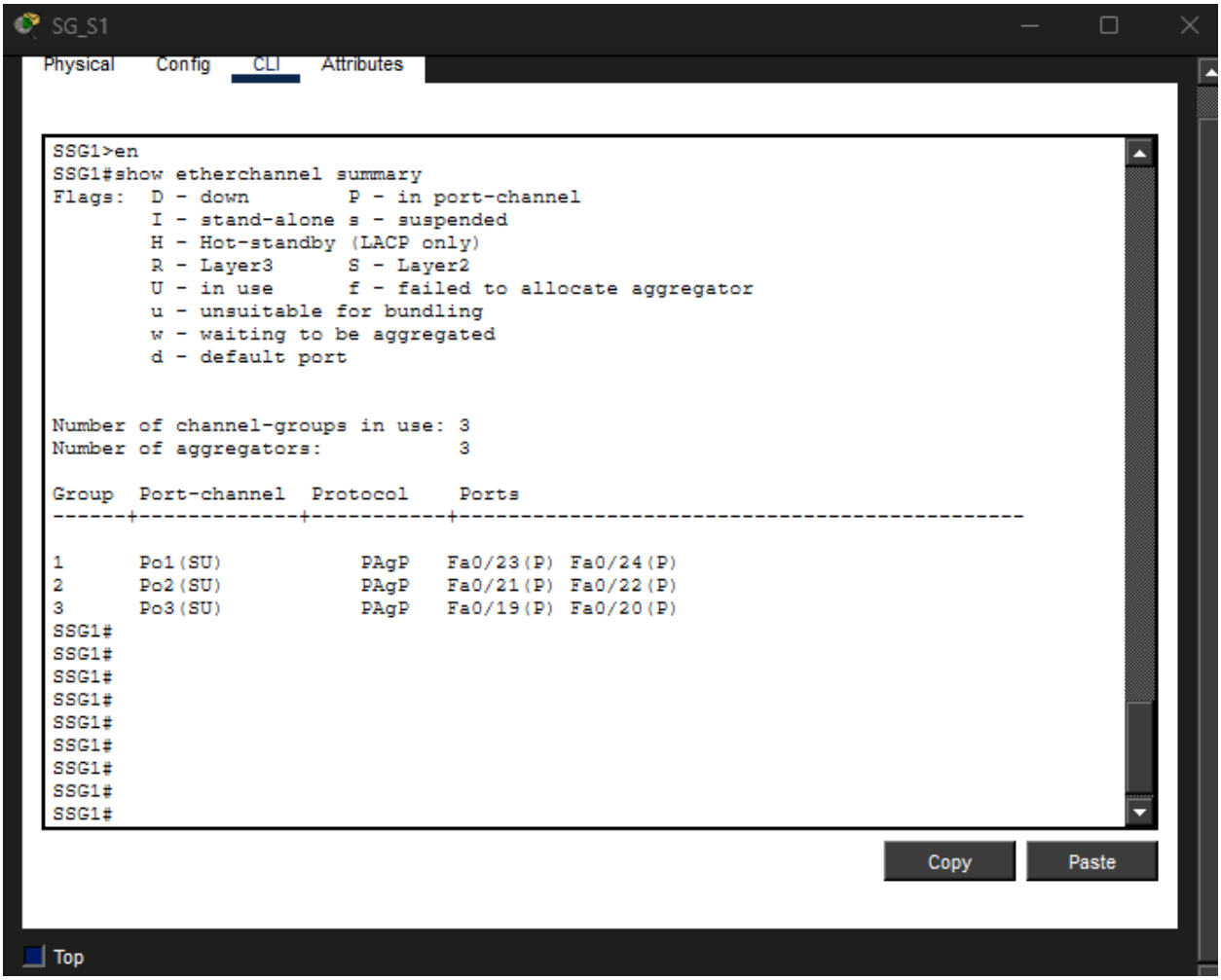
IND_S1

```
SIN1>en
SIN1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

2      Po2(SU)           PAgP   Fa0/21(P) Fa0/22(P)
3      Po3(SU)           PAgP   Fa0/19(P) Fa0/20(P)
SIN1#
SIN1#
SIN1#
SIN1#
SIN1#
SIN1#
SIN1#
SIN1#
SIN1#
SIN1#
SIN1#
SIN1#
SIN1#
```

Copy Paste

Top

SG Switch



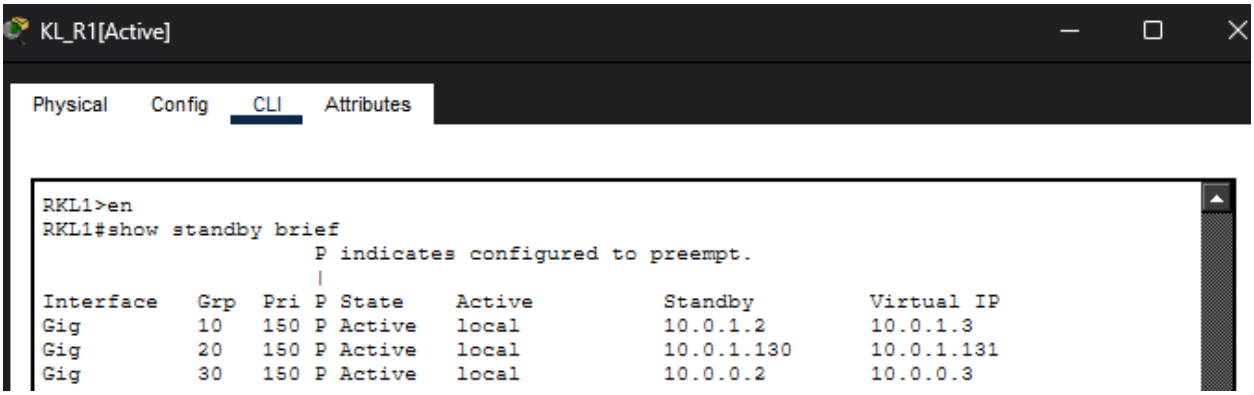
```
SSG1>en
SSG1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 3
Number of aggregators:           3

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           PAgP   Fa0/23(P) Fa0/24(P)
2      Po2(SU)           PAgP   Fa0/21(P) Fa0/22(P)
3      Po3(SU)           PAgP   Fa0/19(P) Fa0/20(P)
SSG1#
SSG1#
SSG1#
SSG1#
SSG1#
SSG1#
SSG1#
SSG1#
SSG1#
```

7.0 HSRP Implementation

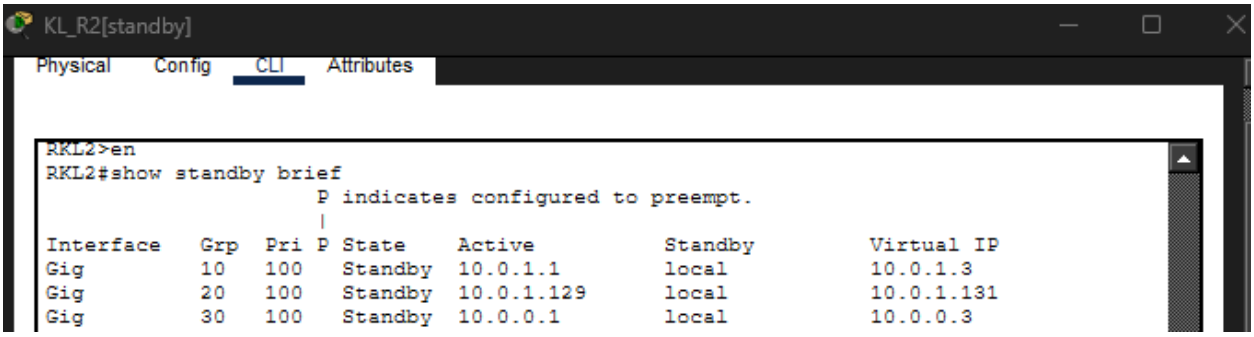
7.1 HSRP Design and Purpose

HSRP is used to provide default-gateway redundancy by presenting a virtual IP and MAC address to hosts. If the active gateway fails, the standby device assumes the virtual identity to maintain gateway continuity with minimal endpoint disruption.

KL Router 1

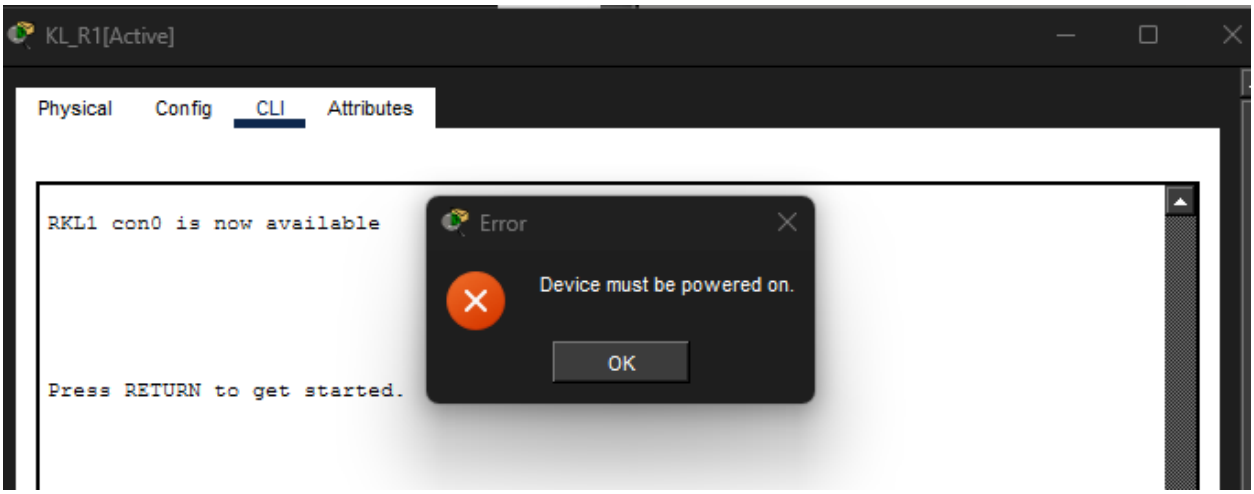


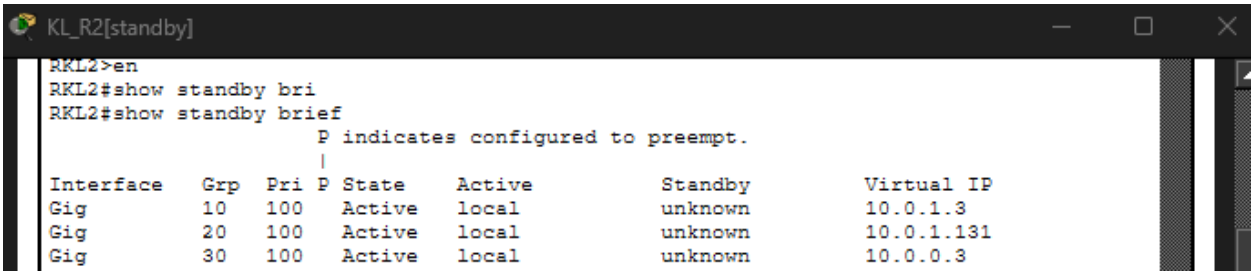
KL Router 2



Router 1 operates as the primary router under normal conditions, while Router 2 serves as a backup. In the event that Router 1 fails, Router 2 will automatically take over to ensure continuous network operation and prevent service disruption.

7.2 HSRP Failover Verification





8.0 Server Farm Verification

8.1 DNS Server

KL Site KL_Management PC1 (VLAN 10) to DNS Server (VLAN 10):

```
C:\>ping 180.30.0.10

Pinging 180.30.0.10 with 32 bytes of data:

Reply from 180.30.0.10: bytes=32 time=6ms TTL=122
Reply from 180.30.0.10: bytes=32 time=19ms TTL=122
Reply from 180.30.0.10: bytes=32 time=5ms TTL=122
Reply from 180.30.0.10: bytes=32 time=5ms TTL=122

Ping statistics for 180.30.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 19ms, Average = 8ms

C:\>tracert www.bungus.org

Tracing route to 180.30.0.11 over a maximum of 30 hops:

  1   5 ms      0 ms      0 ms      10.0.1.1
  2   1 ms      1 ms      1 ms      200.100.100.30
  3   2 ms      1 ms      1 ms      200.100.100.14
  4   2 ms      1 ms      2 ms      200.100.100.6
  5   2 ms      2 ms      1 ms      200.100.100.22
  6   3 ms      4 ms      2 ms      200.100.100.18
  7   2 ms      3 ms      3 ms      180.30.0.11

Trace complete.

C:\>
```

Australia Site Finance_Lap1 (VLAN 20) to DNS Server (VLAN 10):

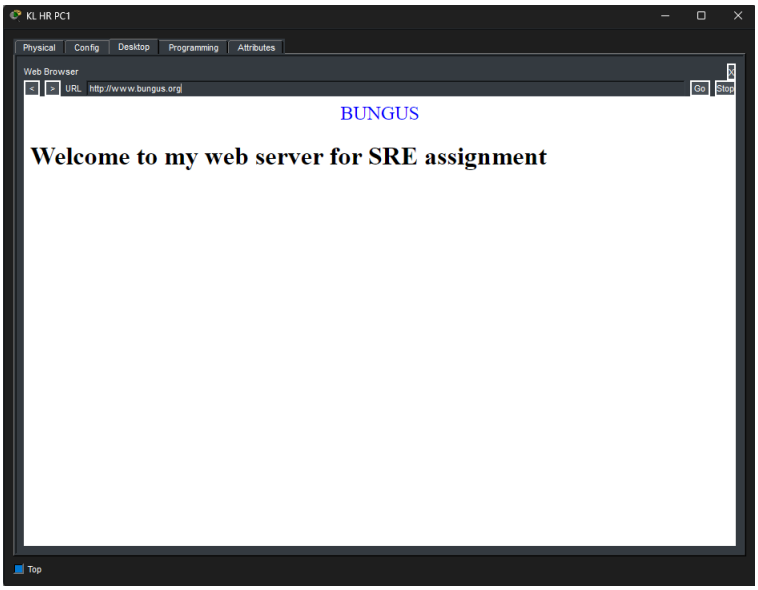
```
C:\>ping 180.30.0.10

Pinging 180.30.0.10 with 32 bytes of data:

Reply from 180.30.0.10: bytes=32 time=30ms TTL=124
Reply from 180.30.0.10: bytes=32 time=12ms TTL=124
Reply from 180.30.0.10: bytes=32 time=67ms TTL=124
Reply from 180.30.0.10: bytes=32 time=23ms TTL=124

Ping statistics for 180.30.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 67ms, Average = 33ms

C:\>tracert www.bungus.org

Tracing route to 180.30.0.11 over a maximum of 30 hops:

  1   7 ms      10 ms     16 ms     172.20.2.1
  2   8 ms      5 ms      6 ms      200.100.100.6
  3   20 ms     5 ms      8 ms      200.100.100.22
  4   15 ms     14 ms     19 ms     200.100.100.18
  5   7 ms      11 ms     8 ms      180.30.0.11

Trace complete.

C:\>
```

Singapore Site SG Management PC1 (VLAN 10) to DNS Server (VLAN 10):

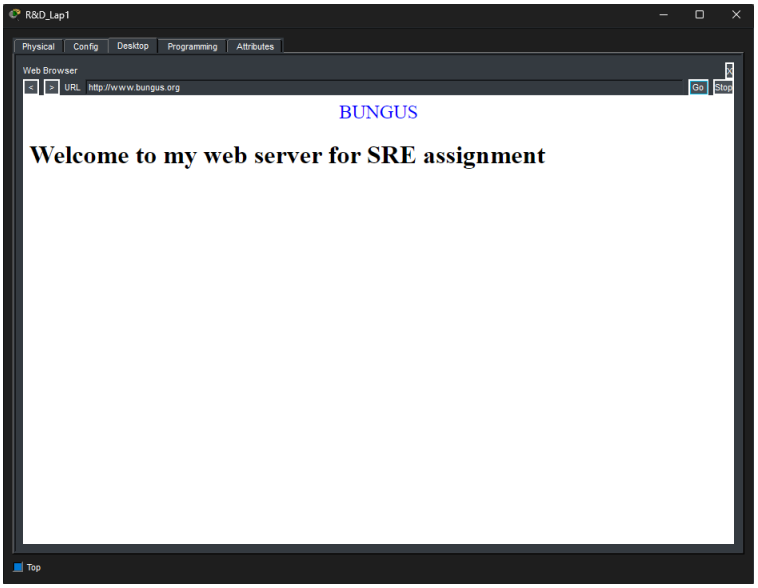
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 180.30.0.10

Pinging 180.30.0.10 with 32 bytes of data:

Reply from 180.30.0.10: bytes=32 time=39ms TTL=124
Reply from 180.30.0.10: bytes=32 time=3ms TTL=124
Reply from 180.30.0.10: bytes=32 time=5ms TTL=124
Reply from 180.30.0.10: bytes=32 time=3ms TTL=124

Ping statistics for 180.30.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 39ms, Average = 12ms

C:\>tracert www.bungus.org

Tracing route to 180.30.0.11 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.10.65
  2   1 ms      0 ms      1 ms      200.100.100.10
  3   30 ms     1 ms      2 ms      200.100.100.22
  4   2 ms      10 ms     1 ms      200.100.100.18
  5   2 ms      2 ms      1 ms      180.30.0.11

Trace complete.

C:\>
```

8.2 Web Server

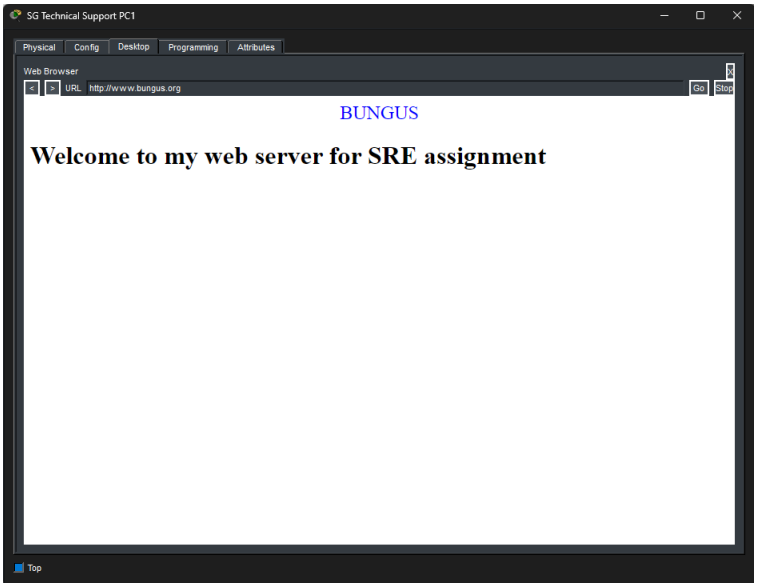
KL Site KL_HR PC1 (VLAN 20) to Web Server (VLAN 10):



Australia Site R&D_Lap1 (VLAN 30) to Web Server (VLAN 10):

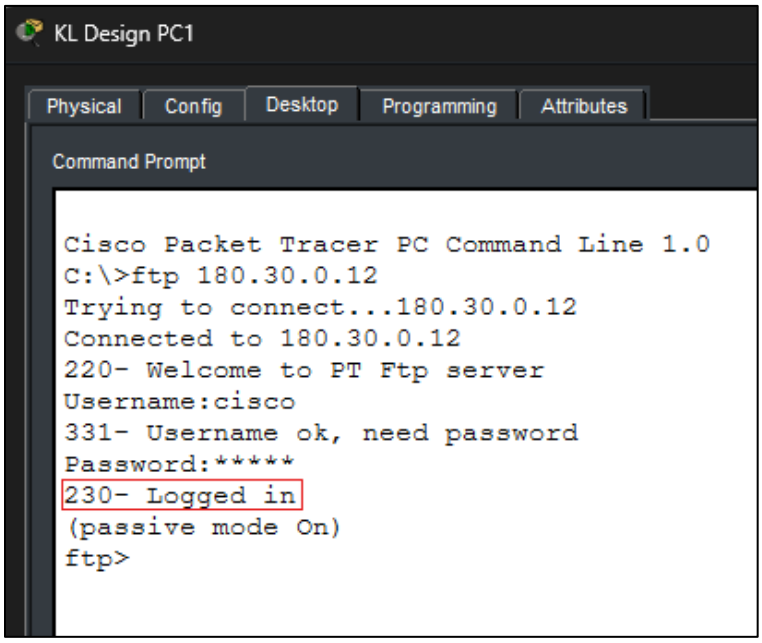


Singapore Site SG Technical Support PC1 (VLAN 40) to Web Server (VLAN 10):

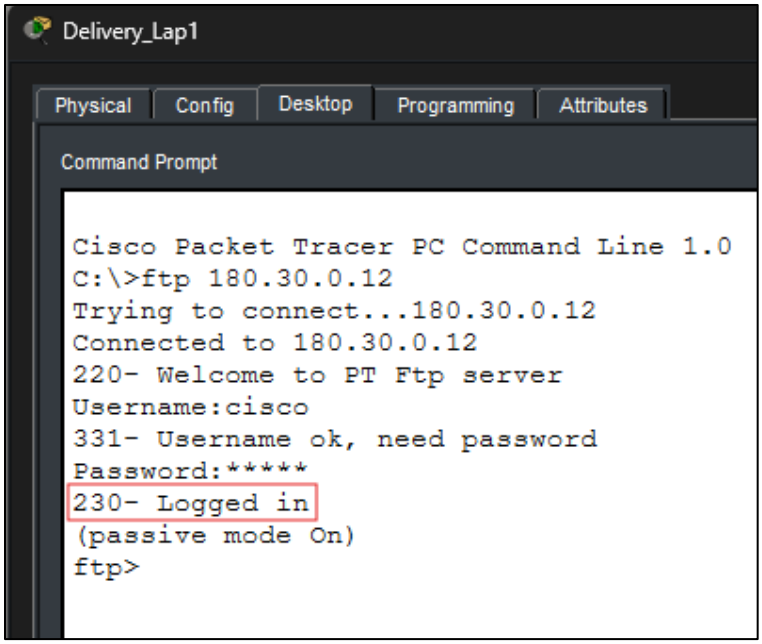


8.3 FTP Server

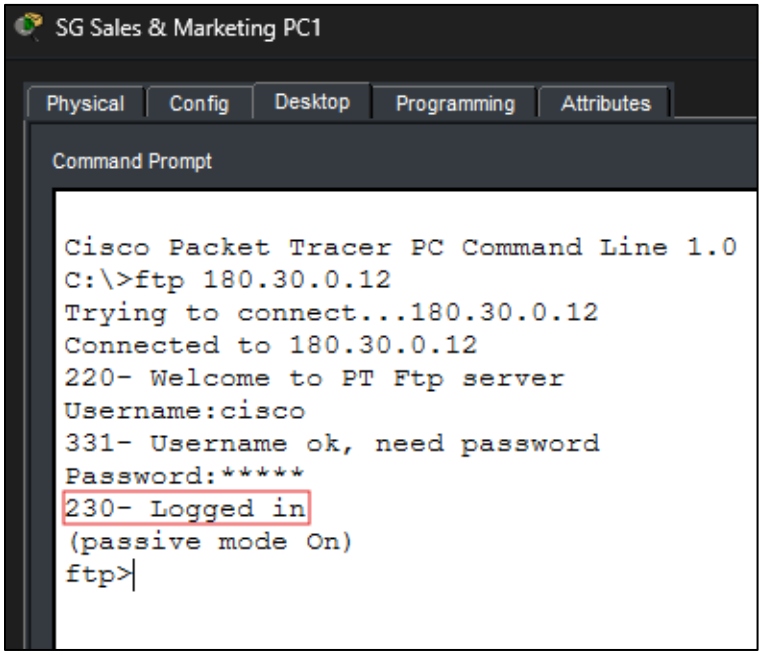
KL Site KL_Design PC1 (VLAN 30) to FTP Server (VLAN 10):



Australia Site Delivery_Lap1 (VLAN 40) to FTP Server (VLAN 10):



Singapore Site SG Sales & Marketing PC1 (VLAN 40) to FTP Server (VLAN 10):



10.0 Layer 2 Security Attacks and Verification

10.1 MAC Address Table Overflow Attack

MAC table overflow attempts to exhaust a switch's CAM table by flooding frames with spoofed source MAC addresses, which may force the switch into increased flooding behavior and raise the likelihood of traffic interception within the VLAN.

Verification:

<u>KL Switch</u>

Switch name: KL_S2

```
interface FastEthernet0/1
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 0000.0C10.A111
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/2
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 00D0.D340.8092
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/3
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 0060.2F74.1185
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/4
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address 00E0.A3BD.6528
 spanning-tree portfast
 spanning-tree bpduguard enable
!
```

AU Switch

Switch name: AUS_S1

```
interface FastEthernet0/1
 switchport access vlan 99
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address 00D0.9788.EC0C
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/2
 switchport access vlan 99
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 0006.2A1B.2459
```

SG Switch

Switch name: SG_S2

```
interface FastEthernet0/1
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 000A.F3A3.D3CE
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/2
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 0060.47C0.8DD7
 spanning-tree portfast
 spanning-tree bpduguard enable
```

IND Switch

Switch name: IND_S2

```
interface FastEthernet0/1
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 0001.438A.B3BC
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/2
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 0060.47AE.12A7
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/3
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 0003.E4C8.975E
 spanning-tree portfast
 spanning-tree bpduguard enable
```

Server Farm Switch

Switch name: Server_S1

```
interface FastEthernet0/1
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 0060.5C88.42A8
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/2
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address 0050.0FCA.8B2E
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/3
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 0060.2F7D.2823
 spanning-tree portfast
 spanning-tree bpduguard enable
```

10.2 DHCP Spoofing Attack

DHCP spoofing occurs when an unauthorized DHCP server responds to clients with malicious or incorrect configuration parameters, while DHCP starvation attempts to deplete the available address pool through repeated lease requests, preventing legitimate clients from obtaining valid leases.

Verification:

```
SAU1#show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
20,30,40,99
Insertion of option 82 is enabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface                    Trusted    Rate limit (pps)
------------------------     -------    ----------------
FastEthernet0/1              no         5
FastEthernet0/2              no         5
FastEthernet0/24             yes        unlimited
FastEthernet0/23             yes        unlimited
GigabitEthernet0/2           yes        unlimited
FastEthernet0/22             yes        unlimited
GigabitEthernet0/1           yes        unlimited
SAU1#
```

10.3 STP Attack

A Spanning Tree Protocol (STP) attack occurs when a malicious or misconfigured device is connected to the network and sends forged Bridge Protocol Data Units (BPDUs) with a lower bridge priority. This causes the attacker's device to be elected as the root bridge, potentially altering the network topology. As a result, legitimate traffic may be redirected, intercepted, or disrupted, leading to network instability, denial of service, or data compromise.

Verification:

```
interface FastEthernet0/1
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 0000.0C10.A111
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/2
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 00D0.D340.8092
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/3
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 0060.2F74.1185
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/4
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address 00E0.A3BD.6528
 spanning-tree portfast
 spanning-tree bpduguard enable
```

10.4 VLAN Hopping Attack

A VLAN hopping attack is a network security attack in which an unauthorized device gains access to traffic from other VLANs. This attack exploits improper switch configuration, allowing an attacker to bypass VLAN separation. Common methods include trunk negotiation abuse and double tagging, where specially crafted frames trick the switch into forwarding traffic to unintended VLANs. As a result, sensitive data from other VLANs may be exposed, compromising network security.

Verification:

```
interface FastEthernet0/1
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 0000.0C10.A111
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/2
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 00D0.D340.8092
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/3
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address sticky 0060.2F74.1185
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/4
 switchport access vlan 10
 ip dhcp snooping limit rate 5
 switchport mode access
 switchport nonegotiate
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 switchport port-security mac-address 00E0.A3BD.6528
 spanning-tree portfast
 spanning-tree bpduguard enable
!
```

11.0 Security Recommendations

11.1 Honeypot Deployment

Security teams can watch harmful activity and gather threat intelligence in a controlled environment by using honeypots, which are purposefully weak or phony systems meant to draw attackers. Since practically every interaction is suspicious, they offer high-quality telemetry with

minimal noise, which aids in testing incident response and verifying detection criteria. High-interaction systems produce rich forensic data at the expense of increased operational risk, whilst low-interaction counterparts provide simple detection. Organizations must use rigorous network segmentation, unchangeable logging, and ongoing maintenance to keep these decoys credible and stop them from serving as launching pads for more assaults. Honeypots are useful early warning systems that improve an organization's overall defensive posture when combined with larger security systems like a SIEM. (Robert, 2025)

A honeypot is a recommended addition to the network design to act as a decoy system intended to attract and monitor potential malicious activity within the simulation. By placing these systems in a controlled and isolated network segment, such as near the India Server Farm or the KL_Design VLAN, the network executive can gather data on how an attacker might attempt to move laterally through the topology. Because this system is not a real production device, any traffic detected interacting with it is a high-fidelity indicator of a security breach or unauthorized internal scanning. This implementation provides a proactive layer of defense, allowing the administrator to identify threats within the Cisco Packet Tracer environment before they can impact critical resources.

11.2 Bastion Host Implementation

A "jump box," or bastion host, is a type of specialized server that serves as a secure gateway between a private subnet and an external network. It serves as a reliable bridge or proxy, enabling administrators to connect remotely via SSH or RDP to private instances without public IP addresses. Businesses use bastion hosts to give isolated environments, like a Virtual Private Cloud (VPC), a secure communication connection without exposing private internal services to the internet. In order to log into instances within the private subnet using secure key pairs, a host must first be launched in a public subnet and then act as a relay. (Nayak, 2021)

It is advised to use a bastion host, sometimes referred to as a leap host, as a hardened entry point for all administrative access to the network infrastructure.
All management traffic should be limited to this one secure server rather than permitting SSH ac cess to every router and switch from any random workstation in the Singapore or KL branches.

Because they will only allow management connections from a certain, trusted IP address, this red uces the attack surface of the core network equipment.
This enhances the network prototype's overall accountability since it allows for the consolidation and monitoring of all administrative changes via a single access point, guaranteeing the security of the management plane across all sites.

11.3 Network Monitoring and Logging

System logs are collections of individual data that give administrators vital insight into system operations, enabling them to spot malicious activity and comply with legal requirements. Organizations should employ Security Information and Event Management (SIEM) solutions for automated monitoring in order to normalize data and identify anomalies because the volume of modern log data can be too much for manual examination. In order to minimize false alarms, effective programs rely on established baselines of typical activity and prioritize recording security-relevant events, such as login attempts and the use of privileged commands. (Dunham, 2019)

It is advised to put in place a centralized network monitoring and logging system to guarantee the Alstom network's high availability and security. The network executive can gather and examine real-time event logs from each router, switch, and WLC in the global topology by setting up a Syslog server and making use of the Simple Network Management Protocol (SNMP). Anomalous behavior, such as repeated unsuccessful login attempts or odd traffic spikes, which could point to a security problem like a DDoS attack, can be detected thanks to this centralized method. In the context of this project, centralized logging guarantees that the student can confirm that all security procedures are operating as intended throughout the branches in Kuala Lumpur, Australia, Singapore, and India, offering total visibility into the infrastructure's condition.

12. Conclusion

The Packet Tracer implementation demonstrates a feasible multi-site enterprise network design for Alstom across Kuala Lumpur (HQ), Australia, Singapore, and India, validated through configuration outputs and connectivity testing. The design applies structured VLAN segmentation with explicit trunking controls to improve manageability and reduce unnecessary Layer 2 exposure across the switching fabric. Inter-site routing is supported through OSPF, enabling scalable route propagation across the WAN, with a default route approach applied where required to maintain consistent reachability.

From a security standpoint, the solution implements a defensible Layer 2 posture by applying port security, DHCP snooping, DAI, and VLAN-hopping controls to mitigate common switching-layer threats. Centralized server farm services (DNS, Web, and FTP) strengthen service availability and provide practical validation targets for end-to-end testing. Finally, additional security recommendations honeypot deployment, bastion host access control, and centralized monitoring/logging provide a practical pathway to improve detection capability, administrative governance, and operational visibility.

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